Labor Market Impacts of Flooding in the United States *

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Abstract

We estimate the sub-annual labor market impacts of over 160,000 flash and slower-onset flood events, using quarterly, county-level employment and wage data for the contiguous U.S. between 1996 and 2023. An additional flash flood day in a quarter reduces county employment by 0.13% and wages by 0.15%. We show that these sub-annual disruptions get masked in annual aggregates. Nationwide flood-induced wage losses average $6.2 billion per year. Suggestive evidence points to physical capital destruction and reduced labor demand as important mechanisms. Our findings inform policies that build labor market resilience against flood shocks, especially in economically vulnerable counties.

Keywords: Flash Floods, Floods, Natural Hazards, Labor Markets, Employment, Wages
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1 Introduction

Flood events are one of the deadliest and most costly forms of natural disasters in the U.S., claiming an average of more than 100 lives and causing billions of dollars in damages annually (Davenport et al., 2021; NOAA National Severe Storms Laboratory, 2024; Congressional Budget Office, 2024). Studies project future increases in flood magnitude and frequency and a higher population exposed to flooding (AECOM, 2013; Jia et al., 2025; Peralta and Scott, 2024). Therefore, it is increasingly important to quantify the full economic consequences of flood hazards, including labor market impacts. The federal government allocates substantial resources to flood-affected regions through emergency assistance, recovery and adaptation programs. For example, in 2022 alone, federal appropriations to reduce flood risks totaled $15 billion (Congressional Budget Office, 2024). Allocating these public funds efficiently requires understanding how floods impact regional economies. Yet existing flood research has primarily focused on annual flood impacts, on large federally declared emergencies, or on urban and coastal settings.

Recent literature using annual data finds that flood events reduce annual output with no contemporaneous effect on annual employment, a pattern interpreted as a productivity shock (Jia et al., 2025). However, this interpretation relies on annual aggregates that may obscure sub-annual labor market disruptions that are either offset through within-year reallocation, concealed by noise in the annual data, or attenuated by averaging when the disruption is confined to a sub-annual period. Such intra-year disruptions could be economically meaningful. Recent studies of other natural hazards document labor market disruptions from hurricanes at both sub-annual and annual frequencies (Hancevic and Sandoval, 2025; Barattieri et al., 2023; Groen et al., 2020; Deryugina et al., 2018; Belasen and Polachek, 2008), from wildfire and wildfire smoke exposure (Coulombe and Rao, 2025; Borgschulte et al., 2024), and from extreme temperature shocks (Garcia et al., 2026). However, there is a lack of evidence documenting the sub-annual impacts of flooding for the contiguous U.S. Importantly, Roth Tran and Wilson (2025) show that while improvements in local capital and productivity after certain hazards such as hurricanes and tornadoes can raise wages and personal income, this is not typically the case for flooding in the U.S. Additionally, beyond the natural hazard research, recent labor market research documents substantial month-to-month income instability driven largely by firm-led demand fluctuations, and such instability is masked in annual data widely used in the literature (Ganong et al., 2025).

In this paper, we study the *sub-annual* labor market impact of flooding for the contiguous U.S.[1] We leverage daily flood exposure data, linked with quarterly, county-level labor market data spanning nearly three decades. We test for differences in impacts by flood type (flash floods versus slower-onset, "non-flash" floods[2]), sector (goods producing versus service producing), location

[1] This includes the 48 contiguous states in the U.S., as well as the District of Columbia (DC).

[2] The SED classifies flash floods as rapid water rises, typically occurring "within minutes to a few hours", representing more sudden, violent upsurges. On the other hand, documented non-flash flood events are classified as those cases where the rise of water is slower and more gradual, typically "over hours to days".

(inland versus coastal counties), and local economic conditions (high versus low unemployment).

We construct a county–quarter panel by combining employment and wage data from the Quarterly Census of Employment and Wages (QCEW), with corresponding flood exposure information from the National Oceanic and Atmospheric Administration's (NOAA's) Storm Events Database (SED). The data span 1996–2023 and include information on over 160,000 flash and non-flash flood events. We use event dates to measure daily, county-level flood exposure, then aggregate to quarterly exposure levels to match the labor market data. Unlike earlier research studying large federally-declared emergencies (Kocornik-Mina et al., 2020; Jia et al., 2025), the SED's comprehensive coverage allows us to capture a wider breadth of flooding events – from spatially and temporally limited incidents to large federally-declared flood disasters. We also include events across the contiguous U.S., unlike previous research which focuses on flood impacts in urban and coastal areas (Kocornik-Mina et al., 2020; Hallegatte et al., 2013) (see figure A1).

Our empirical strategy uses a specification with county-level fixed effects to account for unobserved time-invariant confounders that may affect exposure to flood events and economic outcomes simultaneously. We also include spatially varying time controls, such as state-quarter-year fixed effects, to account for state-specific seasonality and macroeconomic shocks that could coincide with flood exposure. Identification relies on within-county variation, after accounting for common state-quarter-year factors that affect all counties within a state.

Understanding sub-annual effects of flooding is critical for both policy design and assessing impacts at the household level. Policy responses to productivity shocks, such as capital investments or business recovery loans at the firm level differ fundamentally from those addressing labor market disruptions, which may require household-level relief (e.g., enhanced food aid[3]) and income support to prevent deeper economic distress. This is especially true for vulnerable households and communities who face disproportionate flood impacts (Masozera et al., 2007) and may also lack capacity to absorb transitory income losses.

Conceptually, floods can affect sub-annual employment through labor demand channels such as physical capital destruction and consumer mobility constraints that reduce demand for output; and labor supply channels including worker mobility disruptions, out-migration, and mortality, though recent evidence suggests mortality impacts are concentrated among seniors rather than prime-age workers (Chu et al., 2025).

We find that an additional day of flash flooding reduces county-level private employment by 0.13% and total wages by 0.15%, conditional on contemporaneous average precipitation, temperature, and county and state-quarter-year fixed effects. The effect for slow-onset floods is negative but statistically insignificant, suggesting sudden, violent flash flooding has more severe immediate impacts. The results are robust to several alternative specifications discussed in the robustness section. Counterfactual analysis indicates total wage losses due to flood events averaged

[3]See Food and Nutrition Service Response to Active Disasters.

approximately $6.2 billion annually (in real 2023 USD) over 1996-2023. Importantly, when we aggregate the quarterly data to the annual level, these effects are statistically indistinguishable from zero, consistent with prior literature finding no annual employment effects (Jia et al., 2025). These results suggest that sub-annual labor market disruptions due to flash flooding exist but are masked in annual data by temporal aggregation. Because annual employment is the average of quarterly values, a disruption in a single quarter enters the annual coefficient at roughly one-fourth the quarterly magnitude. Since the precision of the annual specification is largely unchanged, we show that it has insufficient power to distinguish the aggregated annual effect from zero.

We examine potential mechanisms through which floods impact labor market outcomes, including plausible channels of physical capital destruction and demand contraction or mobility constraints by analyzing sectoral and sub-sectoral employment and wage responses, impacts on establishment counts, and wages per worker. Our results reveal that impacts span both goods-producing and service-providing sectors. The differing effects across sectors – a decrease in goods-producing establishments and a decrease in wages per worker in the service-providing sector – suggest that capital destruction and reduced labor demand help drive our main results. The decline in wages per worker in the service sector suggests an intensive margin channel, plausibly through reductions in hours worked among hourly workers. The absence of lagged effects after one quarter suggests that permanent out-migration does not drive immediate flood impacts. Since outcomes that would allow more direct tests of alternative channels (such as migration or hours worked) are not observed at the same frequency as our quarterly labor market data, or are unavailable altogether, our evidence for mechanisms is suggestive.

Beyond the studies already discussed, a growing literature documents the economic consequences of flooding and flood risk, including establishment-level impacts on employment, wages, and adaptation following Hurricane Sandy–induced flooding in New York City (Indaco et al., 2021), mobility and rental impacts from high tide flooding (Lee et al., 2024), bankruptcies and debt following Hurricane Harvey-related flooding in Houston (Billings et al., 2022), housing and property prices (Plough et al., 2026; Georgic and Klaiber, 2022; Hennighausen and Suter, 2020; Beltrán et al., 2018), and agricultural and rural outcomes in developed and developing countries (Hodge, 2021; Kim et al., 2023; Kamble et al., 2024; Parida and Chowdhury, 2021). Firm-level evidence from Europe and Vietnam documents persistent negative effects on asset, sales, capital and employment (Fatica et al., 2024; Zhou and Botzen, 2021), though without distinguishing impacts by flood types.

This paper advances the literature in several ways. First, we document the labor market impacts of flooding at a sub-annual level that previous annual analyses could not identify. Existing flood related research primarily uses data at annual frequencies (Boustan et al., 2020; Jia et al., 2025; Roth Tran and Wilson, 2025; Erda, 2024), but sub-annual frequencies may better capture some labor market outcomes. Our findings indicate that meaningful sub-annual labor market shocks exist, suggesting a combined labor market and productivity shock rather than only a productivity shock (Jia et al., 2025).

Second, we explore the impact of a wider range of flood events than is commonly considered in the natural hazard- and flood-related literature. This literature is skewed towards studying economic impacts for large and federally declared flood disasters in the U.S. (Boustan et al., 2020; Jia et al., 2025; Roth Tran and Wilson, 2025). This is despite the fact that the overwhelming majority of flood events are not recorded as federally declared events, as they are limited in spatial scope and duration. For example, in our sample, 94% of all flash flood days and 80% of all non-flash flood days occur outside of FEMA (Federal Emergency Management Agency)-declared flood emergency periods. Such non-declared events could be disruptive for local economies. Incorporating flood events across a greater range of spatial-temporal intensities avoids treatment misclassification and selection bias, capturing more typical flood experiences rather than analyzing an atypical subset of rare, declared disasters that may not represent more frequent local impacts. We show that flash flood days outside of FEMA-declared flood emergency periods also have economically meaningful labor market impacts that are similar in magnitude to our baseline average effects. We also expand the flood literature by distinguishing between the impacts of flash floods and slower-onset, non-flash floods, as well as examining heterogeneity in the impacts across specific sectors. While Collalti (2024b) and Collalti (2024a) provide important evidence documenting the impacts of flash floods on aggregate economic activity, and firm-level reductions in sales and employment in Central America and the Caribbean, to our knowledge we are the first to explicitly distinguish between flash and non-flash flood impacts on labor markets in the U.S.

Finally, this paper reveals an important avenue through which floods can affect welfare in impacted communities. To contextualize our findings, Davenport et al. (2021) find an annual average of $6.63 billion in property, crop, and other damages (in real 2020 USD over 1988-2017) caused by flooding, and Hallegatte et al. (2013) estimate asset-based flood damages at $6 billion USD annually in 2005 for 136 global coastal cities, projected to reach $52 billion by 2050. The significant labor market impacts we find ($>$ $6 billion per year in wage losses) could have welfare implications as labor income and economic activity are disrupted (Hsiang et al., 2017).

2 Background and Data

Drawing on flood events documented in NOAA's Storm Events Database (SED), described in detail below, we construct county-level measures of flash and non-flash flood exposure as the number of days each county experiences flooding within a quarter. Figure A2 shows that almost all counties within the contiguous U.S. have experienced some flood days within the period of 1996-2023. There are numerous inland counties that are exposed to repeated flooding, particularly in the upper Mississippi Valley, Northeast, and Southwest regions. Flash floods are more common in the Southwest and Northeast while slow-onset floods are more common along major rivers (e.g., the Mississippi and Missouri rivers).

While most literature documents flood impacts across urban and coastal areas, figure A1 shows that the intensity of exposure (average days of flooding) is broadly similar across inland and coastal

counties. There are, however, substantially more inland counties than coastal counties, and a greater population in counties exposed to inland flooding compared to coastal floods.

Existing datasets such as the Dartmouth Flood Observatory (Brakenridge, 2025), Emergency Events Database (Delforge et al., 2025), or the FEMA Declared Federal Disasters Database (Federal Emergency Management Agency, 2025) focus primarily on large-scale or federally declared events. Satellite-based datasets (Li et al., 2018), meanwhile, face limitations in detecting flood extent due to issues related to cloud cover and difficulty capturing short-duration flash floods. The SED published by NOAA- National Weather Service (NWS), comprehensively documents a full spectrum of flood events, regardless of magnitude, spatial extent, or duration, while uniquely distinguishing between flash and non-flash floods.

Initially, the SED documented details related to tornadoes, thunderstorms, wind, and hail events, but starting in 1996 the SED expanded significantly to include a comprehensive record of all types of extreme weather-related hazards that are capable of causing fatalities, injuries, property damage, and disruptions to commerce (National Weather Service, 2016). The SED incorporates details related to a wide range of intense meteorological events, including but not limited to, thunderstorms, flash floods, floods, winter storms, hurricanes, droughts, wildfire events, and details of many other weather-related hazards. Each record contains detailed information about event start and end time and dates, geographic coordinates, detailed event and episode narratives, fatalities and property damage estimates in dollars.

The SED is primarily administered and verified by the NWS. However, the comprehensive nature of the SED data also relies on a diverse network of information sources beyond the NWS, including county, state and federal emergency management officials, local law enforcement officials, trained spotters, NWS damage surveys, newspaper clipping services, the insurance industry, and the general public (National Weather Service, 2016). This comprehensive approach therefore helps capture the full scope of extreme weather-related hazards that impact different geographies and communities within the U.S.

One of the advantages of using the SED for studying the economic impact of flooding in the U.S. is that the SED distinguishes between the nature of flooding based on onset patterns. The database differentiates between inland flash floods, which are characterized by sudden and rapid onset with a shorter duration (few hours)[4], and inland non-flash floods, which have a slower or more gradual onset, lasting for longer periods of time (can last for multiple days)[5]. Note that this analysis only

[4]The SED classifies inland flash flooding as a "life-threatening, rapid rise of water into a normally dry area beginning within minutes to multiple hours of the causative event (e.g. intense rainfall, dam failure, ice jam)" (National Weather Service, 2016).

[5]In the SED, inland non-flash floods are documented as "Flood" under event type. The SED classifies inland non-flash floods as any "high flow, overflow, or inundation by water which cause damage. In general, this would mean the inundation of a normally dry area caused by an increased water level in an established watercourse, or ponding of water, that poses a threat to life or property." (National Weather Service, 2016). Additionally, low impact flooding such as "minor flooding in urban areas", "minor ponding of water during or after a heavy rain event or flood", "high stream levels due to steady or slowly rising/receding creeks/streams that do not pose a threat to life or

considers the impacts of *inland* flash flooding, and *inland* non-flash flooding events; the SED also documents coastal flooding which we do not consider[6]. We use the event start and end dates to count the number of days in each quarter that a county experienced flash and non-flash floods[7].

As an alternative to the SED, Gourley et al. (2013) outlines alternative flash flood exposure measures. For example, one approach uses USGS discharge measurements to classify flash flood events as those where water level at a given gauge rises beyond a given flood stage threshold within 6 hours of the onset. While this approach would capture fluvial flooding, it would miss pluvial floods, such as urban flash floods in areas far from a river or stream. Another approach to measuring flood extent uses georeferenced bounding polygons from NWS storm reports, available starting in 2007, to delineate impacted areas. However, not all events have a corresponding polygon and the limited temporal coverage would limit our analysis that begins in 1996. Given these limitations, we use the SED-based flash and non-flash flood events assigned at the county level as our primary measure of exposure. This also maps naturally to our labor market outcomes, measured at the county level, which we discuss next.

Our main dependent variables are labor market outcomes, including employment counts and wages paid. Sub-annual data on employment and wages come from the Quarterly Census of Employment (QCEW). The QCEW provides county-level employment data at both monthly and quarterly frequencies. It also provides the data disaggregated by industry. Employment is the count of filled temporary or permanent jobs, whether full or part time, by place of work[8]. The quarterly reports include each establishment's monthly employment levels for the pay period that includes the 12th of the month (Quarterly Census of Employment and Wages, 2025). The database is built using the federal and state unemployment insurance programs, and is supplemented with information from additional sources. The QCEW data are not estimates, and the Bureau of Labor Statistics (BLS) reports that the QCEW covers 95% of total U.S. employment

property." (National Weather Service, 2016) – all are considered as non-flash inland flood events. An ongoing flood event may intensify into short-term flash flooding with intense rainfall resulting in sudden, rapid increase in flood waters. Conversely, flash flood events may transition into lower-intensity floods as sudden, violent water surges weaken. In such cases, the National Weather Service data preparers use their professional judgment to determine when an event ceases to be classified as flash flood and transitions to a flood classification (National Weather Service, 2016).

[6]The SED provides documentation of coastal flooding, defined as "those portions of coastal land zones (coastal county/parish) adjacent to the waters, bays, and estuaries of the oceans"; further detailed as "Flooding of coastal areas due to the vertical rise above normal water level caused by strong, persistent onshore wind, high astronomical tide, and/or low atmospheric pressure, resulting in damage, erosion, flooding, fatalities, or injuries." (National Weather Service, 2016). The data preparers determine coastal and inland areas where events are classified as Flash Flood or Flood- rather than Coastal Flooding (National Weather Service, 2016). To the extent that concurrent coastal flooding may influence our results- we include all other contemporaneous natural hazards as an additional control in our analysis.

[7]Between 1996 and 2006, for a subset of events, the Storm Events Database (SED) reports hazard exposure at the NWS Public Forecast Zone level rather than at the county level. We map these zones to counties using zone–county crosswalks (National Weather Service, 2025); however, for approximately 0.6% of all flood-related events, a reliable mapping cannot be established because the corresponding zone codes are no longer in use.

[8]Temp agency workers used in construction or healthcare are classified in NAICS 56132. Independent contractors are not included in QCEW. Workers that travel or work away from the employer of record county are reported in the employer county.

(Autor et al., 2024; U.S. Bureau of Labor Statistics (BLS), 2023).

The QCEW reports data on total wages paid within a calendar quarter, i.e. quarterly wage bills, regardless of when the services were performed. Wages include total compensation paid, including bonuses, stock options, severance pay, profit distributions, the cash value of meals and lodging, tips and other gratuities, and, in some states, employer contributions to certain deferred compensation plans (such as 401(k) plans) (Quarterly Census of Employment and Wages, 2025).

Other Data Sources

GridMET- Precipitation and Temperature Data: Quarterly average daily precipitation and temperature for each county in a given quarter-year are used as control variables in the empirical analysis. GridMET provides daily high resolution (4km) surface meteorological data for the US from 1979 onward (Abatzoglou, 2013). We use daily precipitation and temperature data from 1996 to 2023 to align with our sample period of analysis. For our empirical analysis, the daily gridded weather data are averaged at the county-quarter level. Specifically, we compute county-area–weighted daily precipitation and temperature using sub-cell coverage weights and then average these county-level daily values across all days within each quarter.

NOAA Coastal Shoreline County Classifications: The NOAA Coastal Shoreline County classification (NOAA Office for Coastal Management, 2025) identifies the coastal counties that are adjacent to open ocean, major estuaries, and the Great Lakes. Such counties, due to their proximity to water bodies, face significant vulnerability to coastal hazards. FEMA defines a coastal county as one that borders the open ocean or Great Lakes coasts, or, one that contains velocity zones (V-zones)[9] or coastal high hazard areas (NOAA Office for Coastal Management, 2025).

Bureau of Labor Statistics- Local Area Unemployment Statistics: The Bureau of Labor Statistics provides historical local area unemployment statistics. These estimates are derived from multiple sources which include the Current Population Survey, Current Employment Statistics survey, the Quarterly Census of Employment and Wages, various Census Bureau programs and unemployment insurance claims data from multiple state workforce agencies (U.S. Bureau of Labor Statistics, 2025). This data source provides monthly labor force and unemployment statistics dating back to 1990 at the county level, which includes the total number of people employed and unemployed and the not-seasonally adjusted unemployment rate. We use the monthly, not-seasonally adjusted unemployment rate to obtain the average quarterly unemployment rate. We use this as a measure of local labor market conditions prior to any flood exposure.

Small Area Income and Poverty Estimates (SAIPE) on Poverty and Median Household Income: The U.S. Census Bureau's SAIPE provides annual, model-based state and county estimates of poverty and median household income (U.S. Census Bureau, 2022). We use the data from 1997-2023,

[9] "V-zones are areas where wave heights more than 3 feet and/or high velocity water can cause structural damage in a 100-year flood, a flood with a 1-percent chance of occurring or being exceeded in a given year." (NOAA Office for Coastal Management, 2025)

and the estimates for percent of people of all ages in poverty, as well as the median household income. These are estimates and not direct counts; they are produced by combining survey data with population estimates and administrative records (Bell et al., 2007).

The summary statistics of the employment, total wages, inland flash flood days, inland non-flash flood days, average precipitation and temperature are given in table A1. In our data, the average number of flood days per quarter is 0.235 for flash floods and 0.626 for non-flash floods. Conditional on counties experiencing at least 1 flood day in a quarter, the number of days per quarter that a county experiences a flash flood or non-flash flood is 1.681 and 6.659 days, respectively.

3 Empirical Strategy

We analyze how county-level private employment and total wages respond to contemporaneous exposure to flash and non-flash floods using a county-level fixed effects estimation strategy to account for unobserved time-invariant confounders that may simultaneously affect both flood exposure and labor market outcomes, thereby using within-county variation in exposure to flash floods and floods over time. Specifically, we estimate the following baseline specification:

$$\begin{aligned}\log(Y_{cqt}) = \alpha + \beta_1 FlashFlood_{cqt} + \beta_2 NonFlashFlood_{cqt} + \\ \beta_3 AvgPrec_{cqt} + \beta_4 AvgTemp_{cqt} + \lambda_c + \sigma_{sqt} + \epsilon_{cqt}\end{aligned} \tag{1}$$

Here $\log(Y_{cqt})$ is the logarithmic transformation of our outcomes of interest - private employment or total wages for county c in quarter q and year t. We also estimate specifications that analyze the impact of flash floods and non-flash floods on sectoral and sub-sectoral level employment and wages. Our key explanatory variables of interest are $FlashFlood_{cqt}$ and $NonFlashFlood_{cqt}$ which measure the total number of flash flood and non-flash flood or slower onset flood days, respectively, experienced by county c, in quarter q and year t. To isolate the effect of flooding from broader weather patterns, we control for contemporaneous average precipitation- $AvgPrec_{cqt}$ and average temperature- $AvgTemp_{cqt}$ in a county c, in a given quarter q and year t. Rainfall is one of the primary determinants of flooding but can also affect labor market outcomes independent of flooding through disruptions to outdoor work, reduced consumer mobility, and economic activity. Controlling for average precipitation therefore removes potential confounding variation from weather patterns that are correlated with both flood probability and economic activity simultaneously. As a robustness test, we further control for days exceeding various extreme precipitation thresholds to verify our results are not driven by extreme rainfall episodes that coincide with flood exposure. λ_c and σ_{sqt} are county and state-quarter-year fixed effects. State-quarter-year fixed effects account for any state-specific policy changes, or regional economic shocks that affect all counties within a state, and state-specific seasonal employment patterns that vary across quarters and years (e.g., some states report 401k contributions). By including the state-quarter-year fixed effects we can ensure that the flash and non-flash flood estimates are not biased by broader state-level economic shocks or policy

interventions that coincide with contemporaneous flood exposures. We provide additional results with several other alternate specifications in the robustness section. Standard errors are clustered at the county level.

The coefficients of interest are β_1 and β_2, which respectively measure the impact of an additional flash flood day and a slower onset/non-flash flood day on labor market outcomes. Our main identification strategy assumes – conditional on county, state-quarter-year fixed effects, and other controls such as average precipitation and temperature – the incidence of contemporaneous flash and non-flash floods is exogenous to our measures of labor market outcomes, i.e. employment and wages. In summary, we compare the same county across multiple different quarters over time, with and without exposure to flash floods and non-flash floods. As long as such flood events are unanticipated, $\hat{\beta}_1$ and $\hat{\beta}_2$ capture the causal effect of flash flood and non-flash flood exposure on local labor market outcomes. Finally, we use a distributed lag specification with leads and lags to capture the dynamic response of labor markets to flood exposure and to test for differences in trends prior to flood exposure[10].

4 Results

Our main findings reveal that flash floods generate significant sub-annual disruptions to the labor market, while slow-onset floods show minimal impacts. Specifically, we find that an additional day of flash flooding reduces private employment and total wages by 0.13% and 0.15% respectively (column (1) and (5) of table 1). In contrast, an additional non-flash flood day has a smaller and statistically insignificant negative effect on employment and wages (table 1). These results largely hold after controlling for contemporaneous or concurring natural hazards[11] within a given county (columns (2) and (6) of table 1). The nature of flooding matters in determining economic impacts. Flash floods, which are characterized by sudden, rapid and violent onset and with high-velocity flows, are found to have a statistically significant disruption to local labor market outcomes.

The quarterly results provide evidence of sub-annual labor market shocks from flash flood exposure. Using aggregated annual data on employment and wages, both flash and non-flash flood exposure show negative but statistically insignificant effects on private employment and wages (columns (3), (4), and (7) and (8) of table 1). This finding aligns with the results in Jia et al. (2025), who find no significant employment effects from flood events, though their analysis focuses exclusively on large flood events.

[10] In our setting, we do not adopt an event-study style design because flood exposure is a repeated, time-varying treatment that switches on and off across quarters within the same county.

[11] These would include natural hazards such as Heat, Thunderstorms, Frost/Freeze, Drought, Debris Flow, Avalanche and many other events, see National Weather Service (2016) for natural hazards recorded in the SED. Additionally, since recent evidence shows thunderstorms can also have negative labor market consequences (Coronese et al., 2025), we control specifically for contemporaneous thunderstorm days in a county (table A2). Our results largely hold and are consistent with the results in table 1.

« Table 1 about here »

In a separate specification, we estimate a distributed lag model that includes two leads and lagged exposures for the previous three quarters. The lead coefficients at -2 and -1 are close to zero and statistically insignificant for both employment and wages, providing no evidence of differential pre-trends or anticipatory labor market responses prior to flood exposure. This supports our identifying assumption that flash and non-flash flood days experienced in a quarter are unanticipated (figure 1). We find that lagged flash and non-flash floods have statistically insignificant effects on employment (figure 1). However, flash flood exposure in the past one-quarter has a negative and significant effect on total wages (figure 1), suggesting some persistence in lost wages. No statistically significant positive effects after a flood mean that wage losses are not compensated for in future quarters.

« Figure 1 about here »

Overall, these results reveal that flash floods generate sharp short-lived disruptions in local labor markets. Employment and total wages fall significantly in the contemporaneous quarter, reflecting immediate impacts either due to physical capital damages, workplace closures and interruptions, lower labor demand due to a fall in consumption demand, or labor supply disruptions. Figure 1 also suggests that out-migration due to flood exposure is not likely a significant channel. Flood-induced displacement would cause persistent reductions in labor supply, which could push wages up and have more sustained negative effects on employment in subsequent quarters. The absence of such persistence and effects in both employment and wages is inconsistent with significant out-migration as a key mechanism in this case[12]. The effect of the one quarter lagged exposure on contemporaneous employment is negative but statistically insignificant, and by the second quarter lag, coefficient point estimates are close to zero and statistically insignificant. This dynamic pattern suggests adjustment of the labor market, returning employment and wages to their pre-shock levels. However, the recovery is restorative rather than expansionary, i.e. without overshooting prior levels. While prior research documents that some natural hazards, such as hurricanes and tornadoes, can raise wages through post-disaster improvements in local capital and productivity, this has not been found for flood hazards (Roth Tran and Wilson, 2025). Roth Tran and Wilson (2025) mention that this may be due to reluctance to undertake investments in improving local capital stock due to higher perceived probability of repeated flooding, thus limiting the capital upgrading that drives long-run gains following other disaster types.

The short-lived nature of adjustment explains why the annual specifications detect no statistically significant labor market effects. Annual employment is the average of four quarterly employment

[12]We do not directly test for the migration mechanism, primarily because migration data at the sub-annual county level are not available. Some related studies use Internal Revenue Service tax filing-based migration inflow and outflow data at the county-annual level to test for migration impacts (Borgschulte et al., 2024; Coulombe and Rao, 2025).

values, so a disruption in a single quarter enters the annual coefficient at roughly one-fourth the quarterly magnitude; the same factor applies to log wages, since a flooded quarter accounts for approximately one-fourth of the annual wage bill. We use quarterly estimates to construct the annual effect that is implied by the quarterly result. Specifically, we use the weighted sum of the contemporaneous, lead, and lag quarterly coefficients from the distributed lag model represented in figure 1. The weighting of each lead and lag coefficient is based on the mean number of flash flood days in the corresponding quarter, such that all impacts within the same calendar year are counted[13]. The resulting implied annual estimate is -0.038 for employment and -0.063 for total wages, against annual estimates of -0.050 and -0.081 (table 1). These implied estimates fall within the annual confidence interval, which also contains zero (figure 2). The annual specification has power of 12% to 17% against these values (figure 2), and the smallest absolute effect it could detect with 80% power is 0.136 for employment (using annual standard error, 0.049) and 0.173 for wages (using annual standard error, 0.062)- these are roughly three to four times the implied magnitudes.

Figure 2 shows that the annual specification has little power to distinguish the alternate target estimate from zero, and that this reflects attenuation of the coefficient rather than any loss of precision- annual standard errors are marginally smaller (columns (3), (4), (7), and (8) of table 1), and residual treatment variation in the annual data conditional on fixed effects is roughly double that in the quarterly data, which offsets the drop in observations at the annual level. The absence of statistically significant annual effects is therefore not informative of the sub-annual disruption. Had the sub-annual effects persisted largely across four quarters, the annual coefficient would be close to the quarterly estimate, and so the absence of persistence beyond roughly one quarter is what makes the annual data unsuitable for detecting the effect. By exploiting quarterly variation, we uncover labor market disruptions that annual aggregation obscures.

« Figure 2 about here »

It is important to note that the drop in the total wage bill reflects a combined effect of employment reductions *and* potential decreases in hours worked. While QCEW data does not separately report hours, it is plausible that part-time or hourly workers experience reduced hours due to flood related disruptions. Recent literature using payroll records shows substantial sub-annual earnings volatility driven by firm-led labor demand fluctuations, that is more pronounced among labor dependent on hourly work (Ganong et al., 2025). Importantly, this volatility is largely masked in the annual aggregates (Ganong et al., 2025). In this context, our findings suggest that natural hazards such as flash floods represent an exogenous source of similar short-run earnings instability,

[13]Two other alternative approaches give similar results- dividing the contemporaneous quarterly coefficient by 4 yields an implied annual effect of -0.032 for employment and -0.038 for wages; dividing the unweighted sum of the contemporaneous, lead and lag coefficients (from figure 1) by 4 leads to target estimates of -0.028 and -0.058 for employment and wages, respectively. Power against these values ranges from 8% to 15%, and all the implied effects fall inside the 95% confidence interval from the annual specification.

further underscoring the importance of using a sub-annual temporal scale for understanding labor market impacts to localized shocks.

We quantify total lost wages from flooding with a counterfactual exercise, using the combined effect of both flash and non-flash floods. Because the outcome variable is the log of the quarterly county wage bill, the coefficients are semi-elasticities. From the specification, we use the estimates $\hat{\beta}_1$ and $\hat{\beta}_2$ to compute $\hat{\beta}_1 \cdot FlashFloodDays + \hat{\beta}_2 \cdot NonFlashFloodDays$ and obtain the impact on the county-quarter wage bill, i.e., we use the estimates in column (5) of table 1 for Flash and Non-Flash Flood and use the observed number of flash and non-flash flood days in each county-quarter to construct the counterfactual (hazard days set to zero) and sum the implied dollar losses to the year (see figure 3)[14]. Averaging across years yields an average annual loss of about $6.2 billion (2023 dollars). Confidence intervals for the annual and average losses are obtained using the delta method. Since this exercise relies solely on contemporaneous coefficients and does not incorporate lagged effects from the previous quarters, the estimated dollar losses likely represent a lower bound estimate[15].

« Figure 3 about here »

These estimates should be interpreted as partial-equilibrium, short-run impacts. This exercise assumes linearity in the relationship between flood exposure and wage losses, and does not account for any potential spatial reallocation across counties. However, our analysis at the commuting-zone level indicates similar negative impacts on total wages (table A9). Moreover, figure 1 shows no compensating wage gains in subsequent quarters, indicating the estimated losses are not offset by subsequent recovery.

5 Potential Mechanisms

Floods could impact labor market outcomes through labor demand and labor supply channels. We hypothesize two avenues through which floods decrease the demand for labor. Floods can destroy physical capital, leading to establishment closures or limiting operations, and thereby reducing labor demand. Second, floods can limit consumer mobility, causing demand for output to fall in local

[14] The counterfactual wage bill for county c in quarter q in year t is the wage bill that would have been absent flooding. We construct this by exponentiating the negative of the estimated log impact of flooding: $-(\hat{\beta}_1 \cdot FlashFloodDays_{cqt} + \hat{\beta}_2 \cdot NonFlashFloodDays_{cqt})$ and multiplying by the observed wage bill: $W^{cf}_{cqt} = W^{obs}_{cqt} \times \exp(-(\hat{\beta}_1 \cdot FlashFloodDays_{cqt} + \hat{\beta}_2 \cdot NonFlashFloodDays_{cqt}))$. The implied dollar loss for county c in quarter q and year t is then: $Loss_{cqt} = W^{cf}_{cqt} - W^{obs}_{cqt}$. We aggregate $Loss_{cqt}$ across all county-quarters within each calendar year to obtain annual dollar losses, and average across years to obtain the average annual loss.

[15] For the case of flash floods, our preferred estimate implies that one additional day of flash flooding in a quarter reduces the local wage bill by 0.15% (column (5) of table 1). Applying this effect to the observed exposure of flash floods in each county-quarter, we construct counterfactual wages with the hazard turned-off or under a no flash flood scenario, compute the implied losses, and then aggregate to the calendar year. Averaging across years in our estimation sample yields average annual wage losses of $5.9 billion, expressed in 2023 dollars (see figure A3).

consumer-facing sectors. Notably, employment can fall even when establishments are not closed entirely, specifically due to demand contraction, and the wage bill may also be affected through a drop in hours worked due to lower consumer demand. Physical capital destruction is therefore not a necessary condition for labor market disruption. Alternatively, the labor supply channel would operate through worker mobility constraints, out-migration due to displacement, and mortality impacts. We use sectoral and sub-sectoral employment and wage responses, establishment counts, and wages per worker to explore possible mechanisms driving our main results. Sectors differ in exposure pathways, dependence on daily activity, and sensitivity to physical disruption. Sectors tied to physical capital and on-site labor activity (e.g. manufacturing and construction) are likely to face shutdowns when facilities are inundated or damaged, whereas consumer-facing service sectors may face consumer mobility constraints and therefore demand contraction (e.g. sectors such as leisure and hospitality, and retail trade). We then discuss evidence for labor supply channels, assessing whether they are dominant channels in our case, based on the dynamic patterns observed in the empirical results and evidence from past literature.

Our sectoral analysis reveals that an additional day of flash flooding reduces employment in goods-producing sectors by 0.27% (column (1) of table 2). The corresponding effect on service-providing sectors is a smaller and statistically insignificant reduction of 0.08% (column (2) of table 2). The capital intensity of goods-producing sectors suggests that capital destruction plays a role in driving our main results. An additional flash flood day reduces total wages in goods-producing and service-providing sectors by 0.29% and 0.14% (columns (3) and (4) of table 2), respectively. Slow onset inland floods have small and statistically insignificant effects across both sectors, further reinforcing that high-velocity, sudden, and rapid onset flash flooding causes more labor market disruption. The service sector experiences a statistically significant decrease in wages per worker (see column (8) of table 2) along with a statistically insignificant decline in employment, suggesting that the flood-induced reduction in worker hours accounts for a meaningful share of the service-sector wage bill decline rather than employment separation alone. While QCEW data does not separately report hours worked, future research could leverage payroll or administrative records to directly quantify the change in hours worked.

We estimate a model with establishment counts as the dependent variable and find evidence that flash floods reduce goods-producing establishments (see column (5) of table 2). The decline in the relatively more capital intensive goods-producing establishments following flash flood exposures provides further evidence of the physical capital destruction channel.

« Table 2 about here »

We further examine sub-sectoral impacts in industries that face distinct vulnerability channels. Specifically, we focus on trade, transportation and utilities (and retail trade within this), leisure and hospitality, and construction. Sectors reliant on consumer mobility and day-to-day demand (retail, leisure and hospitality) experience meaningful reductions in employment and wage bills following flash floods (table A3). These effects are consistent with temporary constraints to access

and consumer mobility, reductions in consumption demand, and disruptions to foot traffic. This provides evidence that flood-driven changes in consumer demand also impact local labor markets.

Interestingly, we find negative and significant impacts of both flash floods and non-flash floods on construction-sector employment and total wages (table A3). This suggests that even low-intensity flooding with gradual onset can disrupt labor activity in the construction sector through prolonged exposure to inundation and work delays. The construction sector's reliance on outdoor work sites makes it vulnerable to flooding that creates waterlogging, regardless of flood intensity or speed of onset.

In terms of the labor supply channels – if flooding led to permanent out-migration and displacement and if such supply effects dominate, wages per worker should rise and not fall and there would also be more sustained negative effects on employment in subsequent quarters. In our case flood exposure lowers wages per worker and we do not find any evidence of such persistence in employment and wages (see figure 1). Additionally, recent evidence shows that mortality impacts from similar inland flash flooding and non-flash flooding are concentrated among senior populations (above 65 years old) rather than prime-age workers (Chu et al., 2025). This suggests that mortality, out-migration or displacement led labor supply impacts are not a dominant channel. If temporary worker mobility disruptions were a dominant supply-side channel, we would expect lower wages per worker across sectors because of fewer hours worked. However, the lack of significant decline in wages per worker in the goods-producing sector suggests that existing workers are not working fewer hours. Alternatively, the larger impacts in the service sectors could be argued to be driven by higher concentrations of hourly employees in sub-sectors such as retail and leisure and hospitality (Ganong et al., 2025). Despite this, the goods-producing sector also has a significant share of hourly workers (for example, in 2024, 57% of private goods-producing workers were paid hourly, compared to 54% in services (U.S. Bureau of Labor Statistics, 2025a,b)). Therefore, while we cannot rule out labor supply reductions, our results are consistent with the demand reduction mechanism. Labor demand is most likely reduced by physical capital damage and by lower consumption demand stemming from reduced foot traffic and consumer mobility constraints.

6 Heterogeneity

Next, we examine heterogeneous effects of flood days across location (coastal versus inland counties), and by existing local labor market conditions. This approach identifies which communities face the greatest vulnerability to flood-related labor market disruptions.

Heterogeneous Effect by Coastal County: The existing literature largely focuses on the impacts and economic costs of flooding in coastal areas or coastal cities (Hallegatte et al., 2013; Strauss et al., 2015; Desmet et al., 2015, 2018), with comparatively less emphasis on *inland* flooding. Some recent literature does study the economic impacts of large federally declared floods for the contiguous U.S. (Jia et al., 2025; Erda, 2024) but does not distinguish the impacts across inland

and coastal counties. Coastal shoreline counties may experience greater vulnerability to flood-related disruptions than inland counties for multiple reasons. Coastal counties are far more densely populated and economically concentrated. The population density in coastal shoreline counties is over six times greater than inland counties (NOAA, 2013), and so a given disruption can affect more workers, establishments and infrastructure. Finally, disruptions could be more likely as it can be compounded over multiple repeated natural hazards such as storm surge, high tides and sea level rise which potentially lead to larger damages and result in a prolonged impact. Specifically, we examine the heterogeneous effects of inland flooding (flash and non-flash) across geographically inland and coastal counties; we do not examine the impacts of coastal flooding events.

Flooding in inland counties is not rare or negligible (see figure A2). As discussed before, there are numerous inland counties in the U.S. that experience flood events over multiple days on average during a given year. Moreover, we have discussed about the extent of flood exposure across inland and coastal counties (see figure A1). This makes the coastal-inland county comparison empirically meaningful for understanding the heterogeneous labor market effects of flooding.

We identify the location-based heterogeneity by interacting our flash flood and non-flash flood days exposure measure with a coastal county indicator based on the NOAA Coastal Shoreline County classification (NOAA Office for Coastal Management, 2025). Our indicator variable takes a value of 1 if the county is a coastal shoreline county, and 0 otherwise.

« Table 3 about here »

We find that an additional day of flash flooding reduces employment and wages in inland counties by 0.11% and 0.13%, and the corresponding effects for coastal counties are larger, at 0.25% and 0.37% respectively. While the point estimates suggest larger coastal effects, the difference in effect size between inland and coastal counties is statistically insignificant (table 3).

Using these estimates in column (2) of table 3, we further quantify total wage losses due to flash and non-flash floods, by inland and coastal counties. We construct counterfactual county-quarter wage bills with flood exposures set to zero and aggregate to calendar years. Figure A4 reports the resulting annual losses by coastal versus inland counties[16]. Average annual losses in wages in inland counties are about 41% of the quantified losses in coastal counties and close to about one-third of all flood related losses to wages.

Heterogeneous Effect by Local Labor Market Conditions: We also test how existing local labor market conditions moderate the impacts of flooding. We measure local labor market strength with a time-variant proxy of labor market slack, defined by the county-level unemployment rates in the preceding quarter relative to the national distribution. Specifically, we classify a county as high-slack if its unemployment rate in the past quarter exceeds the 90th percentile of the national distribution in the same year-quarter. We use the past quarter unemployment rate to identify

[16]The confidence intervals are obtained based on the delta-method with county-clustered standard errors.

high or low slack counties to capture the most recent labor market conditions and because the contemporaneous unemployment rate can interact endogenously with contemporaneous flood exposures. In the appendix, we test for robustness to cumulative flood exposure.

Low slack counties are closer to full employment levels and have stronger underlying demand conditions than in high slack counties. In contrast, high-slack counties are operating below full employment, with weaker labor demand, lower vacancy creation, and reduced private-sector absorption capacity (Shimer, 2005; Imbens and Lynch, 1993). Within the search and matching framework, job finding depends on labor market tightness and empirically high unemployment coincides with lower vacancy creation (Shimer, 2005). Existing evidence also suggests re-employment probabilities for workers are lower in high unemployment areas, thus possibly making flood induced separations harder and costlier to reverse in high-slack places (Imbens and Lynch, 1993). On the other hand, evidence also suggests that vacancy yields (hires per vacancy) decline with labor market tightness and that firms adjust through costly instruments in the form of recruitment intensities, screening, and compensation (Davis et al., 2013), thus, implying replacement costs are higher in tight labor markets which strengthens the incentives to maintain labor rather than separate during transitory shocks. Consequently, upon experiencing contemporaneous flooding, high slack counties could face greater vulnerability to economic and labor market disruptions, whereas low-slack counties are better positioned to buffer the shock through retention or faster reallocation.

Using our definition of labor market slackness, figure A5 maps the spatial distribution of counties ever classified as high-slack during our sample period, shading them by the unemployment rate at the time they crossed the threshold[17]. High-slack counties appear across both inland and coastal regions, with many counties in parts of the midwest, the rural south, the southwest, and certain west coast areas; many of these also experience flash flood and non-flash flood hazards.

« Table 4 about here »

Our results provide evidence that there is a differential impact for both flash and non-flash floods between high slack and low slack counties. An additional flash flood day reduces employment and wages by 0.41% and 0.59% respectively in high-slack counties (columns (1) and (2) of table 4), while the effect for flash flooding in low slack counties indicates a smaller drop in employment and wages by 0.10% and 0.12% respectively (columns (1) and (2) of table 4). These differences are statistically significant (columns (1) and (2) of table 4), confirming that stronger labor markets buffer against severe flood shocks.

We observe similar patterns for lower intensity, slower onset flooding. An additional day of non-flash floods reduces employment and wages by 0.097% and 0.12% respectively in high-slack

[17]For counties classified as high-slack in multiple quarters, the map reports the maximum unemployment rate observed across all high-slack episodes for that county during the sample period, as it is not feasible to display each episode separately.

counties (columns (1) and (2) of table 4), while the effects in low-slack counties are close to zero and statistically insignificant. Even in this case, the difference in effect size between high and low-slack counties is statistically significant, suggesting a differential effect even for slower onset flood events. This is likely because high slack counties have weaker adjustment capacity, and with lower re-employment probabilities, even slow onset non-flash floods can negatively impact employment and total wages.

We also test this heterogeneous effect using alternate definitions of persistent slack-counties, classified as high-slack if unemployment exceeded the 90th percentile in *each* of the past four quarters (columns 3-4 of table 4), or *each* of the past eight quarters (columns 5-6 of table 4). These measures proxy persistently weak economic conditions in recent quarters, and chronic labor market weakness. This measure relies on identifying slack over an extended time period unlike the previous measure which relies on the unemployment rate only in the past quarter. In columns 3-6 of table 4 we find larger negative effects of both flash and non-flash floods on employment and wages for high-slack counties compared to columns 1-2 of table 4, thus suggesting persistent slack leads to larger impact from flooding. These results reinforce that underlying labor market strength plays an important role in shaping labor market vulnerability to floods.

A potential concern with our slack-based heterogeneity design is that the high-slack classification relies on lagged unemployment rates. If previous flood exposures elevated unemployment in affected counties, then our high-slack indicator could partially reflect persistent effects of earlier flood shocks rather than underlying labor market conditions or labor market strength. In this case, the interaction between current flood exposure and the high-slack indicator might conflate genuine heterogeneity with the persistence effects of past disasters. To mitigate such concerns, we further control for lagged and cumulative flood exposures in past quarters. The results are reported in table A4. The estimated coefficients remain very similar to those in table 4, though we observe some modest attenuation in effect size after controlling for cumulative flood exposure, suggesting some possible role of past shocks contributing to the slack classification without altering the overall pattern of heterogeneity. Finally, even after accounting for cumulative past flood exposure, high slack may still proxy for other time-varying local vulnerabilities (e.g., sectoral composition, fiscal capacity, credit constraints, or infrastructure conditions) that can influence baseline flood risk, damages, and the local capacity to recover from flood hazards.

We also test if the heterogeneity by labor market slack drives the difference in effects for inland and coastal counties. For each county, we calculate the share of sample quarters in which it is classified as high-slack under each slack definition, and then compare mean values between coastal and inland counties using a simple difference-in-means test. Table A5 shows no systematic evidence that coastal counties are more frequently classified as high-slack. If anything, under one of the more persistent slack definitions, high-slack status is more prevalent in inland counties. This suggests that heterogeneity in coastal versus inland counties is not driven by differences in labor market conditions.

Overall, the heterogeneous effects align with the labor market literature suggesting low-slack

economies operating closer to full employment are more resilient and potentially have greater absorptive capacity to reallocate displaced workers following disruptions. Existing evidence shows that fiscal stimulus and interventions have stronger effects when unemployment is high and private-sector absorption capacity is limited. Our results suggest that targeted post-disaster federal aid may be most valuable in high-slack counties, where flood induced disruptions generate disproportionate labor market impacts.

We examine heterogeneous effects using alternate measures of economic vulnerability, i.e., county-level poverty rates and median household income from 1997-2023. We classify counties as high-poverty rate if the previous year exceeds the 90th percentile of the national distribution, and as low-median-income if the median household income falls below the 10th percentile[18]. Both measures are time-varying to capture recent relative economic conditions at the county level.

Table A6 shows that flash floods and non-flash floods have a larger negative effect in economically vulnerable counties. Low-median income counties experience significantly larger wage losses from both flash floods (net effect -0.76%) and non-flash floods (-0.13%), as well as significantly larger employment losses from non-flash floods. High poverty counties exhibit significantly larger employment losses from non-flash floods (-0.077%). While the pattern of larger negative effects in vulnerable counties holds across both measures and flood types, some interaction terms are imprecisely estimated and not statistically significant. All specifications include controls for cumulative flash and non-flash flood exposure over the past eight quarters. These patterns reinforce that poor underlying economic conditions amplify flood vulnerability.

7 Robustness

Previously, we discussed the robustness of our results to controlling for other concurring natural hazards and thunderstorms. In this sub-section, we explore the robustness of our results to a range of alternative specifications, controlling for extreme precipitation, conducting the analysis at the commuting zone level, allowing for spatially correlated errors, and considering flood severity. We also conduct a placebo analysis that supports our assumption that additional flood days are exogenous to local labor market conditions. Results of all these analyses are reported in the appendix and discussed below, e.g., results with alternative specifications is presented first below.

Alternative to our main specification, we include county-year fixed effects to capture any county-specific shocks or business cycles. We further include hydrological unit (HUC) subregion (HUC-4)

[18]High Poverty is a time-varying classification based on whether a county's poverty rate in the previous year is above the 90th percentile of the national county-level poverty distribution in the given year (=1), the indicator variable takes value 0 otherwise. Low Median Income is also a time-varying indicator based on whether a county's median household income in the previous year is below the 10th percentile of the national county-level median income distribution in that given year (=1), the indicator variable takes value 0 otherwise. Due to inconsistent county coding for Connecticut in 2022–2023 in the poverty and income related data source (SAIPE), we assign Connecticut counties to 0 for both indicators. This is assumed because Connecticut counties are never classified as high-poverty or low-income in earlier years of the same data.

or basin-level (HUC-6) seasonal fixed effects, i.e., HUC 4-quarter or HUC 6-quarter fixed effects, as robustness checks. Counties within the same basin can share similar hydrological and seasonal flood exposure patterns. Gourley et al. (2013) document a relationship between basin catchment size and flood flashiness. HUC-quarter fixed effects therefore aim to capture such hydrological seasonal patterns, ensuring that the results are not primarily driven by hydrological unit-specific seasonal differences across counties within a given state. Our main results are robust to including county-year fixed effects to capture county-specific shocks as well as hydrological unit-specific seasonal fixed effects (table A7).

To ensure that our results are not driven by periods of extreme rainfall episodes we control for contemporaneous rainfall anomalies by including the number of days a county experiences rainfall exceeding 25.4 mm (or 1 inch), 50 mm, 75 mm or 100mm in a given quarter. While the baseline specification controls for average precipitation, extreme rainfall episodes may not be fully absorbed by this control. Table A8 shows that while the flash flood coefficients attenuate slightly as we control for increasingly severe rainfall thresholds, the negative effect on both employment and total wages remains in a similar range and is statistically significant across specifications.

Previous literature documents labor market dynamics at the CZ level (Autor et al., 2013, 2024; U.S. Department of Agriculture, Economic Research Service, 2025) because labor markets often span multiple counties due to commuting flows and supply-demand linkages. Counties, on the other hand, may not correspond to the functional geography within which labor market adjustments occur. To ensure our results are not driven by the choice of spatial unit, we conduct our analysis at the CZ-level. We obtained the county to commuting zone, and commuting zone to census division crosswalk files from Autor and Dorn (2013). We find largely similar and consistent results, reported in table A9, supporting the robustness of our results.

A potential concern in county-level analyses of natural hazards is that spatially correlated shocks across nearby counties can lead to downward-biased standard errors and spurious statistical significance if not accounted for. To address this, we re-estimate our baseline specification by allowing for spatial clustering based on spatially robust Conley standard errors (Conley, 1999) with varying cutoffs at 25 km, 50 km, 75 km and 100 km, this allows for spatial correlation across nearby counties. Our results are robust and statistically significant after allowing for this spatial correction (table A10).

Table A11 examines whether the baseline average effects are driven by counties experiencing high-intensity flooding (large reported damages) or by counties experiencing periods of Presidentially declared flood disasters. This is important to test because our results could be driven by the most extreme events. To proxy for high intensity floods we use the NOAA Storm Events Database (SED) to compute total reported flood damages in each county–year–quarter, defined as the sum of reported property and crop damages. Importantly, multiple events in SED have zero or missing damage values, but that does not necessarily imply damages are truly zero, as it could be incomplete and under reported. We then construct a High Damage Reported indicator that takes the value 1 if a county's reported damages in a given year-quarter exceeded the 90th percentile

of the positive-damages distribution over 1996–2023 (i.e., conditional on damages being strictly greater than zero), and 0 otherwise. About 8% of all flash flood days and 12% of all non-flash flood days overlap with the high damage quarters. Using this classification, we do not find that the effects are concentrated only in high-damage county-quarters. Columns (1) and (2) of table A11 show the impacts of an additional flash flood day on employment and wages in county-quarters with no or low reported damages are similar in magnitude to the baseline estimates in table 1, and the differences across high- and no-or-low-damage periods are not statistically significant.

Next, we use the FEMA Presidential Disaster Declarations database to identify flood-related emergency declarations. The FEMA Declared Emergency indicator is set to 1 for county-year-quarters that overlap with a Presidentially declared flood disaster period, and 0 otherwise. The results in columns (3)-(4) of table A11 show that additional flash flood days *outside* of Presidentially declared periods also reduce employment and wages. However, the net effect of an additional flash flood day during a Presidentially declared flood disaster is larger (more negative) than during non-declared floods for both employment and total wages, and this difference is statistically significant.

Overall, these findings support the interpretation that the baseline labor market impacts of flash flooding are not an artifact of the extreme, high-damage flooding events. Instead, adverse labor market impacts are visible even in quarters without high reported damages and outside federally declared emergencies, while major FEMA declared emergencies have larger quarterly impacts on employment and wages.

Finally, to ensure that our estimated effects reflect the timing of flood exposure rather than spurious correlation with labor market outcomes, we conduct a within-county placebo analysis. In this, we shuffle flood exposure within each county over time, which preserves the distribution of flash flood exposure within a county while breaking the link between the true timing of flooding and labor market outcomes. We estimate our baseline specification 1000 times on these placebo exposures and construct the distribution of coefficients using the random exposure timing. Our observed estimates lie far in the tails of the placebo distributions (see figure A6) indicating that the effects we observe were unlikely to arise by chance. We also conduct a within state-quarter-year placebo, shuffling flash flood exposures across counties within the same state-quarter-year. This preserves the temporal distribution of flash floods while breaking the spatial link between exposure and outcomes. Our observed estimates again lie at the tails of the placebo distribution (see figure A7), confirming the observed effects are unlikely spurious.

8 Conclusion

In this paper, we examine the labor market impacts of flood hazards. Our findings provide new evidence on the sub-annual labor market impact of flash and non-flash flood exposure, revealing dynamics that complement and extend existing research. While past literature documented productivity shocks from large flood events with evidence of no contemporaneous change in

annual employment (Jia et al., 2025), our sub-annual analysis uncovers significant labor market disruptions at the quarterly level that are not observable over a longer temporal window (i.e., at the annual level), with the sub-annual impacts not made up for in future quarters. The absence of significant annual effects reflects attenuation from mechanical aggregation, rather than an absence of labor market disruption. The annual specification has low power against the effects that the quarterly estimates imply, and the annual confidence intervals contain both those implied values and zero. Past literature documents productivity shocks from large events, while we show that even moderate sized flash floods have short term labor market consequences. These patterns are important to uncover as they matter for documenting losses in economic welfare and for improving rapid policy responses to such weather-related hazards.

In our analyses, we distinguish between the nature of flood events, documenting the impact of inland flash floods which have a more sudden and rapid onset, versus inland non-flash floods which have a more gradual onset yet last longer. This distinction is empirically important as the nature of flooding could determine the differential patterns of economic disruption. We also include floods that range in severity and do not make a distinction in terms of spatial scope and duration, or federal declaration status, providing a more comprehensive assessment of flood impacts than studies that have focused solely on large-federally declared disaster events (e.g., Kocornik-Mina et al., 2020; Jia et al., 2025; Roth Tran and Wilson, 2025). These results suggest that disaster responses may need to consider short-run labor market impacts even for floods that do not rise to the level of a declared emergency.

The sectoral impacts we document provide evidence for mechanisms driving our results. Reduced goods-producing establishments and employment following flash flood exposure suggest physical capital destruction, while the decrease in wages per worker within the service-providing sector is consistent with our hypothesis of reduced hours worked due to lower consumer demand, though disruptions to worker mobility cannot be entirely ruled out. The absence of persistent impacts on employment and wages further indicates that permanent out-migration and mortality are unlikely to be dominant mechanisms. Together, the results are consistent with labor demand fluctuations as the primary driver of flood-induced labor market disruptions.

We also uncover heterogeneous effects by geographic location (coastal versus inland counties) and by local labor market strength. Specifically, we show that although the effect size of non-flash flood and flash flooding in coastal counties is larger, the difference is statistically insignificant and inland counties also suffer a negative effect of flash flooding on employment and wages. Importantly, we also find that pre-flood labor market conditions can shape labor market vulnerability to floods. High labor market slack counties (counties with past-quarter unemployment rates above the 90th percentile of the national distribution) experience a larger negative impact of both flash flooding and non-flash flooding on labor market outcomes than low-slack counties. This finding suggests that flooding can worsen already struggling local economies, highlighting the need for targeted policy interventions based on local economic conditions. It also suggests that investments that create favorable employment conditions can

enhance labor market resilience to floods.

Despite its contributions, our study has some limitations which could be addressed in future research. While using the SED data to measure flood exposure has advantages in terms of its comprehensive coverage, it lacks details related to the exact spatial extent and severity of inundation. Additionally, we assign flood events to the entire county, whereas events could be spatially localized, impacting a relatively small area, and affecting only local populations and establishments. In this regard, future research could leverage high-resolution cloud free satellite data to more precisely map inundation extent and link flood exposure to specific populations and establishments, providing more accurate estimates of economic losses. Additionally, incorporating real-time mobility and spending data could capture short-term loss in economic welfare as well as immediate behavioral and adaptive responses that our quarterly analysis cannot capture. Such analysis could further document the mechanisms through which floods disrupt local economies and inform more effective policy responses.

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9 Tables

Table 1: Effect of Flash Flood and Non-Flash Flood Days on Total Private Employment and Wages

Dep. Vars.:	log(Emp.)				log(Tot.Wages)			
	Quarterly		Annual		Quarterly		Annual	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Flash Flood	-0.126**	-0.124**	-0.050	-0.048	-0.153**	-0.152**	-0.081	-0.080
	(0.053)	(0.053)	(0.049)	(0.049)	(0.065)	(0.065)	(0.062)	(0.062)
Flood	-0.008	-0.008	-0.0004	-0.0003	-0.008	-0.008	-0.002	-0.002
	(0.011)	(0.011)	(0.008)	(0.008)	(0.014)	(0.014)	(0.009)	(0.009)
Other Natural		-0.010**		-0.005***		-0.006		-0.003
Hazards		(0.004)		(0.002)		(0.005)		(0.003)
Observations	345,797	345,797	86,648	86,648	345,797	345,797	86,648	86,648
St×Qtr×Year FE	Yes	Yes			Yes	Yes		
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
St×Year FE			Yes	Yes			Yes	Yes

Notes: Controls include average precipitation and average temperature. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 2: Effect of Flash and Non-Flash Flood Days on Private Employment, Total Wages, Establishments, and Wages Per Worker in Goods-Producing and Service-Providing Industries

Dep. Vars.:	log(Emp.)		log(Tot. Wages)		log(Establishments)		log(Wages Per Worker)	
	Goods-Producing (1)	Service-Providing (2)	Goods-Producing (3)	Service-Providing (4)	Goods-Producing (5)	Service-Providing (6)	Goods-Producing (7)	Service-Providing (8)
Variables								
Flash Flood	-0.272***	-0.075	-0.293***	-0.137**	-0.075*	0.002	-0.020	-0.062**
	(0.092)	(0.050)	(0.104)	(0.060)	(0.045)	(0.043)	(0.038)	(0.027)
Non-Flash Flood	-0.011	-0.007	-0.014	-0.006	0.015	0.005	-0.003	0.0003
	(0.020)	(0.012)	(0.024)	(0.013)	(0.012)	(0.008)	(0.009)	(0.005)
Observations	343,489	344,445	343,489	344,445	343,489	344,445	343,489	344,445
St×Qtr×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (county-level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 3: Heterogeneous Effect by Inland-Coastal Counties

Dependent Variables:	log(Emp.) (1)	log(Tot. Wages) (2)
Variables		
Flash Flood	-0.111**	-0.129*
	(0.056)	(0.069)
Non-Flash Flood	-0.005	-0.006
	(0.012)	(0.015)
Flash Flood X Coastal	-0.143	-0.238
	(0.124)	(0.159)
Non-Flash Flood X Coastal	-0.041	-0.037
	(0.025)	(0.035)
Net Effect- Flash Flood in Coastal Counties	-0.254**	-0.366**
	(0.117)	(0.148)
Net Effect- Non-Flash Flood in Coastal Counties	-0.046**	-0.043
	(0.022)	(0.033)
Observations	345,797	345,797
St×Qtr×Year FE	Yes	Yes
County FE	Yes	Yes

Notes: Indicator variable- Coastal takes 1 if the county is a coastal shoreline county, and 0 otherwise. Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

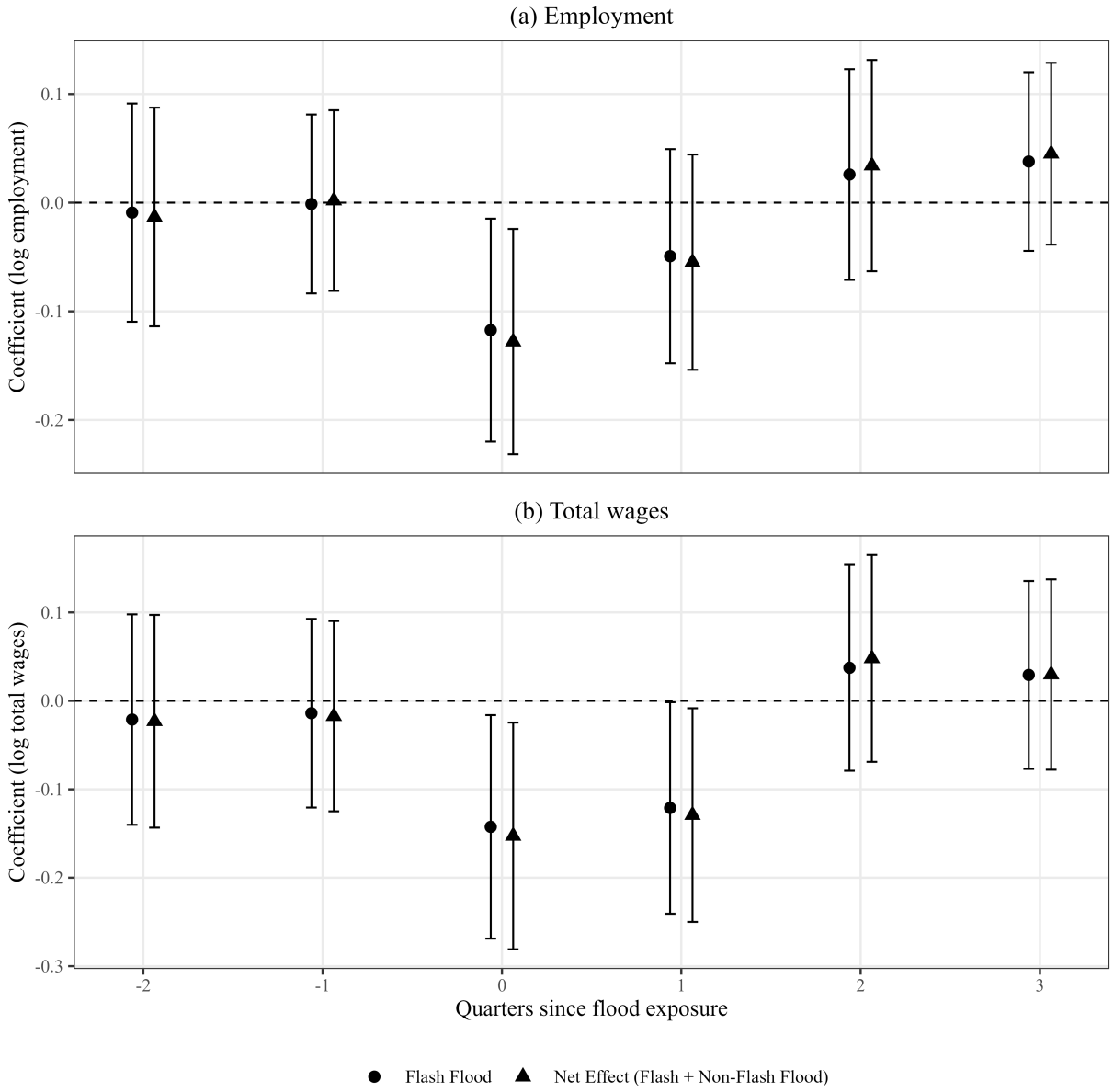
Table 4: Heterogeneous Effect by High-Low Slack Counties (Alternate Definitions)

Dependent Variables:	log(Emp.) (1)	log(Tot.Wages) (2)	log(Emp.) (3)	log(Tot.Wages) (4)	log(Emp.) (5)	log(Tot.Wages) (6)
Variables						
Flash Flood	-0.102*	-0.116*	-0.099*	-0.115*	-0.106**	-0.128*
	(0.053)	(0.064)	(0.053)	(0.065)	(0.054)	(0.066)
Non-Flash Flood	-0.001	-0.0002	-0.0002	0.0003	0.0008	0.001
	(0.012)	(0.015)	(0.012)	(0.015)	(0.012)	(0.015)
Flash Flood × High Slack	-0.308*	-0.474*				
(=1 if Past Qtr. Unemp. > 90th perc.)	(0.177)	(0.246)				
Non-Flash Flood × High Slack	-0.096*	-0.115**				
(=1 if Past Qtr. Unemp. > 90th perc.)	(0.051)	(0.054)				
Flash Flood × High Slack			-0.761***	-1.08***		
(=1 if *all* Past 4 Qtrs. Unemp. > 90th perc.)			(0.250)	(0.356)		
Non-Flash Flood × High Slack			-0.209***	-0.224***		
(=1 if *all* Past 4 Qtrs. Unemp. > 90th perc.)			(0.068)	(0.074)		
Flash Flood × High Slack					-0.864**	-1.06**
(=1 if *all* Past 8 Qtrs. Unemp. > 90th perc.)					(0.338)	(0.480)
Non-Flash Flood × High Slack					-0.303***	-0.322***
(=1 if *all* Past 8 Qtrs. Unemp. > 90th perc.)					(0.073)	(0.089)
Net Effect- Flash Flood in	-0.410**	-0.590**				
High Slack Counties	(0.179)	(0.248)				
Net Effect- Flash Flood in			-0.860***	-1.196***		
High Slack Counties			(0.251)	(0.357)		
Net Effect- Flash Flood in					-0.971***	-1.193**
High Slack Counties					(0.334)	(0.477)
Net Effect- Non-Flash Flood in	-0.097**	-0.115**				
High-Slack Counties	(0.048)	(0.051)				
Net Effect- Non-Flash Flood in			-0.209***	-0.224***		
High-Slack Counties			(0.066)	(0.071)		
Net Effect- Non-Flash Flood in					-0.302***	-0.321***
High-Slack Counties					(0.071)	(0.086)
Observations	345,776	345,776	345,797	345,797	345,797	345,797
St×Qtr×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: In columns (1) and (2) the High Slack indicator variable takes value 1 if in the past quarter the unemployment rate in the county was above the 90th percentile of the national distribution over the same quarter-year, and 0 otherwise. Alternatively for columns (3) and (4), counties are classified as high-slack if unemployment exceeded 90th percentile in each of the past four quarters, and finally in columns (5) and (6) counties are classified as high-slack if unemployment exceeded 90th percentile in each of the past eight quarters. Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

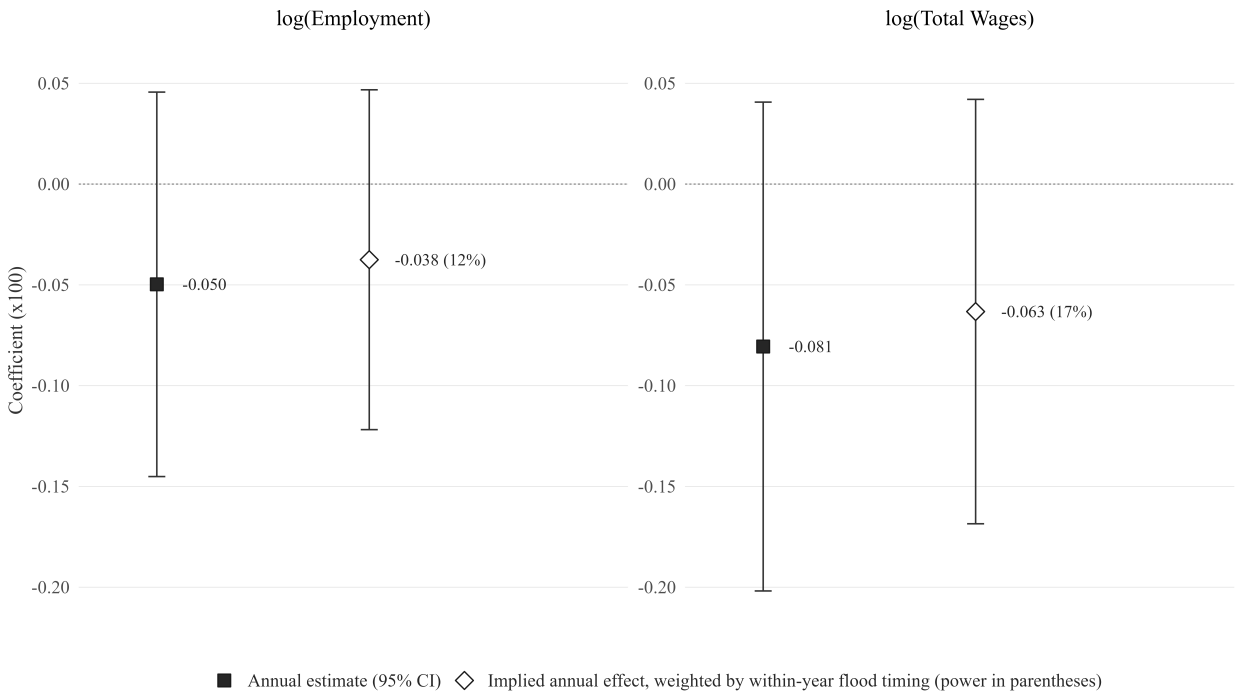
10 Figures

Figure 1: Flood Effects on Employment and Total Wages: Contemporaneous, Lead and Lagged Flood Exposure



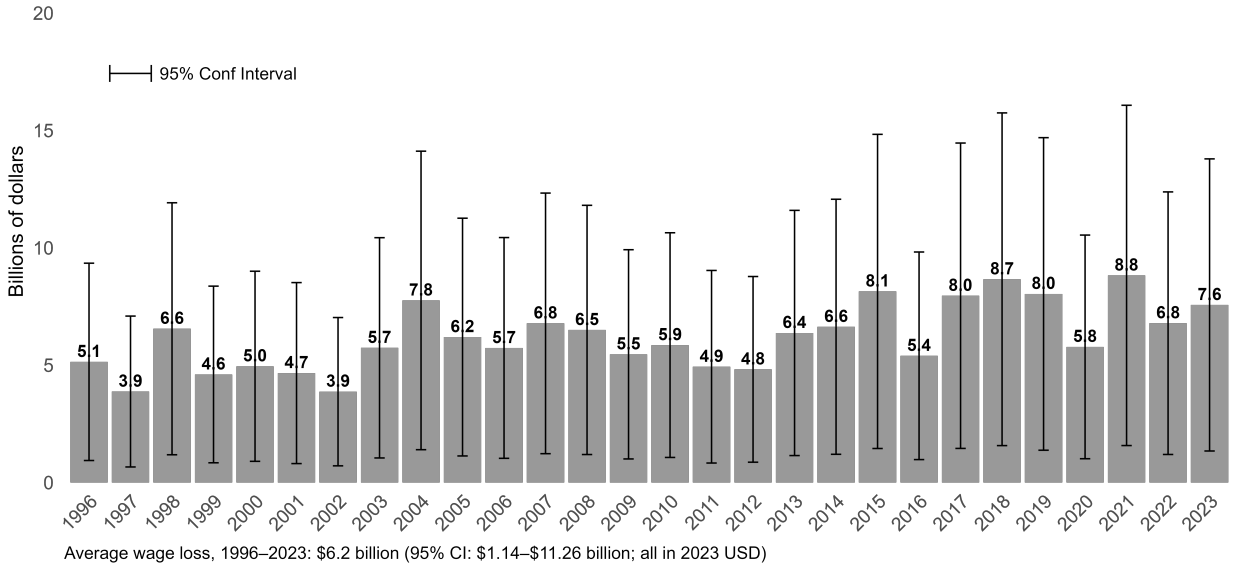
Notes: This plots the contemporaneous, lag and lead coefficients from the distributed lag model using quarterly data. Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects, with clustered (county-level) standard errors. The vertical bars indicate 95% confidence intervals. All coefficients and standard errors have been multiplied by 100.

Figure 2: Power analysis using alternate target estimates



Notes: The filled square is the estimated effect of an additional flash flood day from the annual specification, with its 95% confidence interval. The unfilled diamond is the annual coefficient implied by the quarterly estimates under temporal aggregation; the percentage in parentheses gives the power of the annual specification against this target estimate, at the 5% significance level. Because annual employment is the average of four quarterly values, a disruption confined to one quarter enters the annual coefficient at roughly one-fourth its quarterly magnitude. The same factor applies to log wages, since the flooded quarter accounts for about one-fourth of the annual wage bill. The implied estimate weights each contemporaneous, lead, and lag coefficient from the distributed lag model in figure 1 by the mean number of flash flood days in the quarter of flood occurrence, so that lead and lagged responses are counted only when they fall inside the same calendar year. The minimum effect detectable (in absolute value) with 80% power at the 5% significance level is 0.136 for employment and 0.173 for wages, roughly three times the implied magnitudes. The implied estimates fall inside the annual confidence intervals, which also contain zero. The annual specification therefore cannot distinguish these effects from zero.

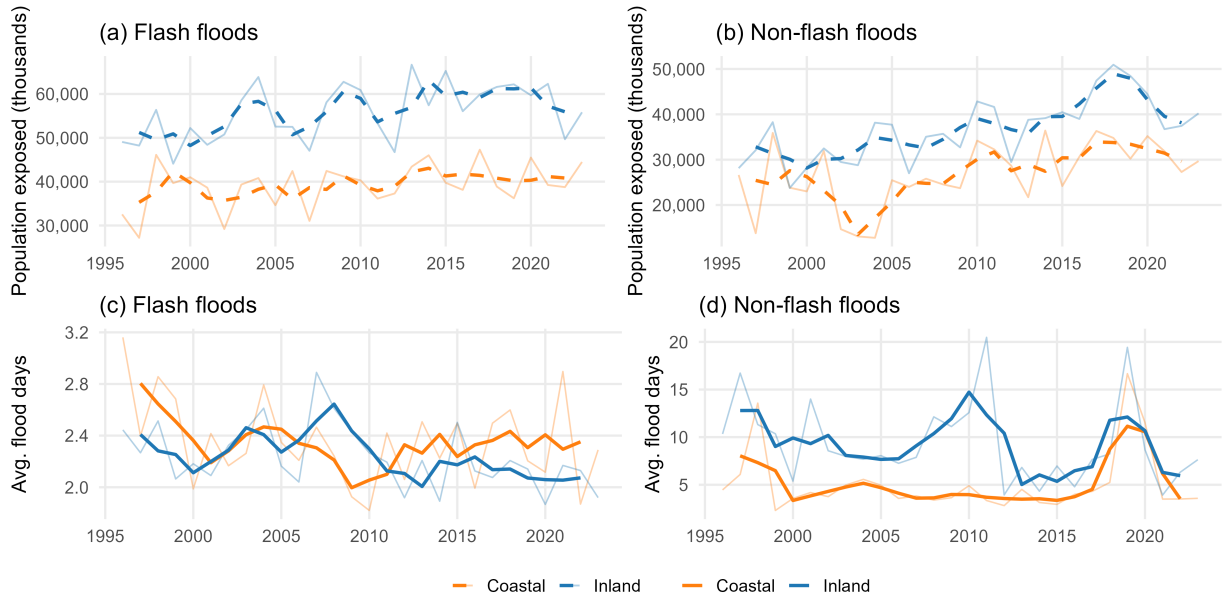
Figure 3: Annual Wage Losses due to Flash Floods and Non-Flash Floods Combined (Real, 2023$)



Notes: Here we use the estimates in column (5) of table 1 for Flash Flood and Non-Flash Flood and use the observed number of flash and non-flash flood days in each county-quarter to construct the counterfactual (hazard days set to zero) and sum the implied dollar losses to the year. The standard errors are computed using the delta method.

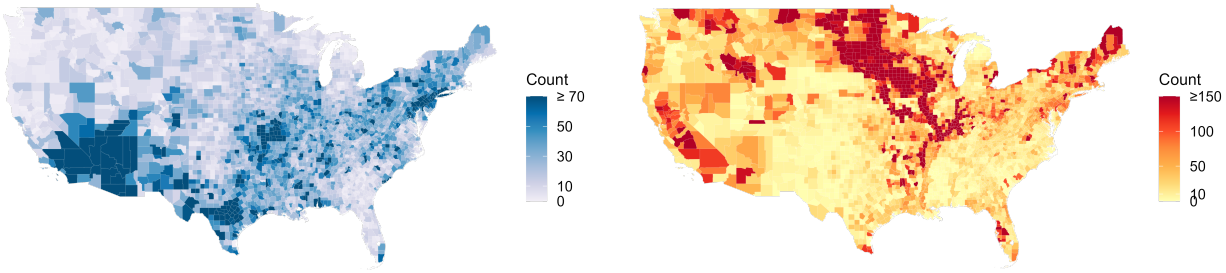
11 Appendix: Figures and Tables

Figure A1: Flash and Non-Flash Flood Exposure in Coastal and Inland Counties



Notes: Panels (a)–(b): Total prime-age population (in thousands, age group 25-64) in affected counties, for flash and non-flash floods respectively. Panels (c)–(d): Average flood days across affected counties, for flash and non-flash floods respectively. Orange = coastal, blue = inland; thin lines = annual, thick lines = 3-year centered moving averages. Note that we include inland flooding in coastal and non-coastal counties but exclude flooding in "those portions of coastal land zones (coastal county/parish) adjacent to the waters, bays, and estuaries of the oceans.", as defined in the SED.

Figure A2: Count of Flash Flood or Non-Flash Flood Days by U.S. County (1996–2023)



(a) Flash floods

(b) Slow-onset / non-flash floods

Table A1: Summary Statistics

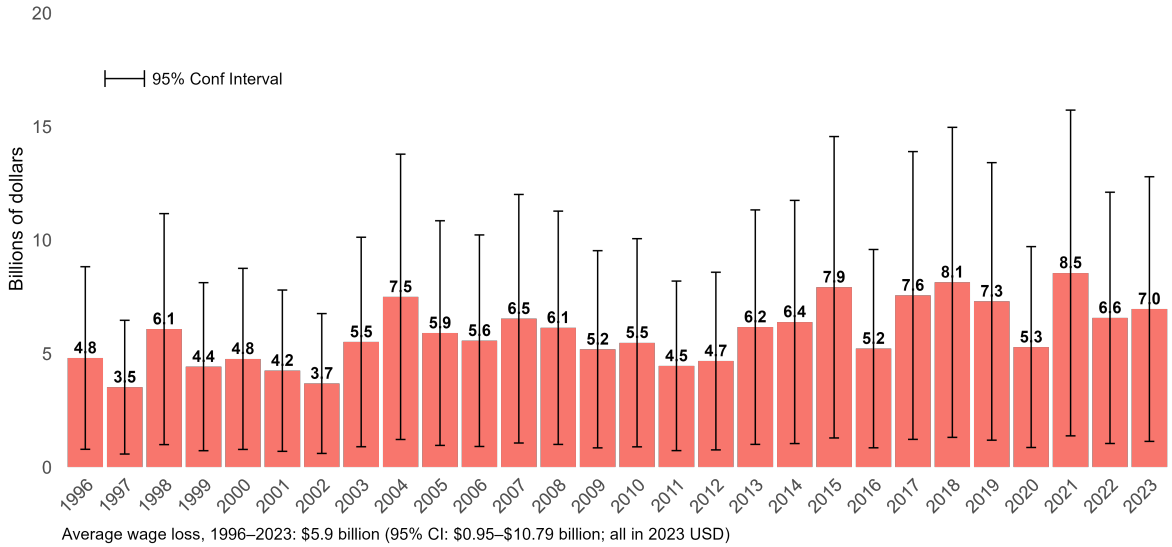
Dependent Variables:	Mean (1)	SD (2)	Sample (3)
Variables-Quarterly			
Employment	35653	130663	345797
log(Employment)	8.879	1.677	345,797
Total Wage (in million USD)	423.6	2,116.9	345,797
log(Total Wages)	17.854	1.837	345,797
Flash Floods	0.235	0.759	345,797
Flash Floods (*Conditional on Flash Flood* > 0)	1.681	1.304	48,247
Non-Flash Floods	0.626	4.199	345,797
Non-Flash Floods (*Conditional on Non-Flash Flood* > 0)	6.659	12.142	32,505
Quarterly Average Daily Precipitation (mm)	2.814	1.614	345,797
Quarterly Average Daily Temperature (C)	12.87	9.058	345,797
Other Natural Hazards	9.089	18.919	345, 797
Other Natural Hazards (*Conditional on Other Natural Hazards* > 0)	11.192	20.426	280,834
Variables-Annual			
Employment	35576	130497	86,648
log(Employment)	8.872	1.684	86,648
Total Wage (in million USD)	1,690.5	8,394.9	86,648
log(Total Wages)	19.233	1.846	86,648
Flash Floods	0.937	1.700	86,648
Conditional on Flash Flood > 0	2.274	1.994	35,707
Non-Flash Floods	2.502	11.176	86,648
Conditional on Non-Flash Flood > 0	9.356	20.073	23,173
Annual Average Daily Precipitation (mm)	2.812	1.147	86,648
Annual Average Daily Temperature (C)	12.86	4.581	86,648

Table A2: Effect of Flash Flood and Non-Flash Flood Days on Total Private Employment and Wages

Dependent Variables:	log(Emp.) (1)	log(Total Wages) (2)
Flash Flood	-0.116**	-0.140**
	(0.053)	(0.065)
Non-Flash Flood	-0.008	-0.008
	(0.011)	(0.014)
Thunderstorm	-0.066	-0.085
	(0.044)	(0.052)
Observations	345,797	345,797
St×Qtr×Year FE	Yes	Yes
County FE	Yes	Yes

Notes: Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure A3: Annual Wage Losses due to Flash Flooding (Real 2023$)



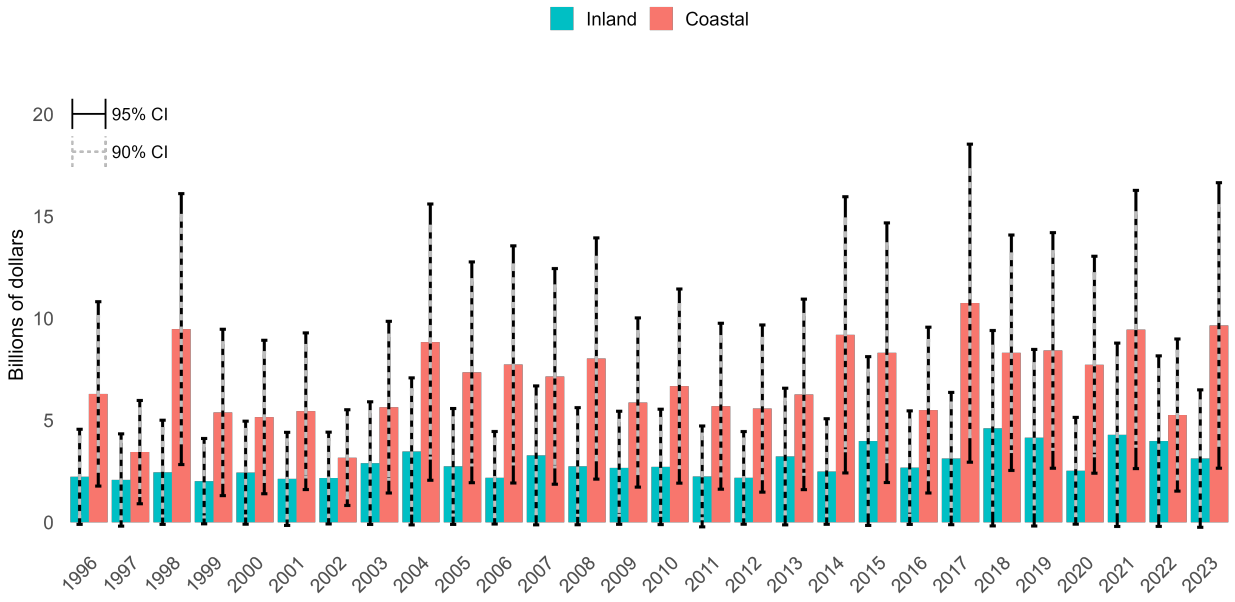
Notes: To calculate the total wage losses, we use our preferred estimate of one additional day of flash flood in a quarter reducing local wage bill by 0.15% (column (5) of table 1). We use this to construct counterfactual wages with the hazard turned-off or under no-flash flood scenario, compute the implied losses and then aggregate to the calendar year. The standard errors are computed using the delta method.

Table A3: Effect of Flash and Non-Flash Floods Days on Private Employment and Wages in Specific Industries

Dep. Vars.:	log(Emp.)				log(Tot.Wages)			
	Trade, Transpt. Utilities	Retail	Leisure & Hospitality	Construction	Trade, Transpt. Utilities	Retail	Leisure & Hospitality	Construction
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Flash Flood	-0.106*	-0.124**	-0.159**	-0.225**	-0.114	-0.126**	-0.246***	-0.441***
	(0.056)	(0.050)	(0.073)	(0.098)	(0.070)	(0.058)	(0.087)	(0.125)
Non-Flash Flood	-0.012	-0.007	-0.023	-0.053**	-0.010	-0.005	-0.033	-0.068**
	(0.019)	(0.014)	(0.020)	(0.025)	(0.020)	(0.016)	(0.025)	(0.030)
Observations	346,157	340,503	337,409	305,721	346,157	340,503	337,409	305,721
St×Qtr×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure A4: Annual wage losses due to Flash Floods + Non-Flash Floods, by Inland and Coastal counties (Real, $ 2023)



Notes: Here we use the flash and non-flash flood estimates in column (2) of table 3 and use the observed number of flash and non-flash flood days in each county-quarter and by inland and coastal counties to construct the counterfactual (hazard days set to zero) and sum the implied dollar losses to the year. The standard errors are computed using the delta method.

Figure A5: High-Slack Counties based on Past Quarter Unemployment Rate

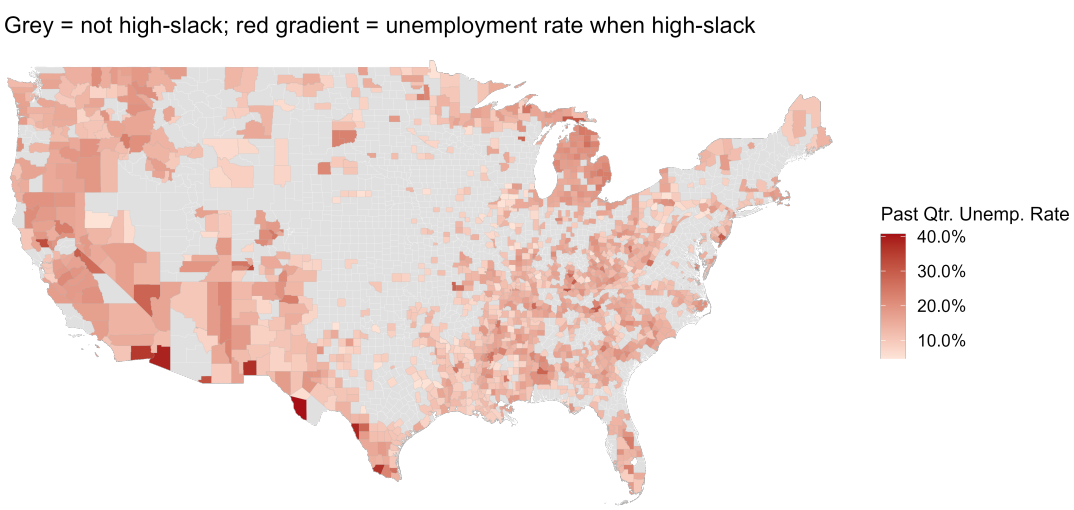


Table A4: Heterogeneous Effect by High-Low Slack Counties

Dependent Variables:	log(Emp.)	log(Tot.Wages)	log(Emp.)	log(Tot.Wages)	log(Emp.)	log(Tot.Wages)
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Flash Flood	-0.111**	-0.126**	-0.107**	-0.116*	-0.119**	-0.138**
	(0.052)	(0.063)	(0.049)	(0.059)	(0.048)	(0.059)
Non-Flash Flood	0.0004	0.001	-0.004	-0.004	0.004	0.005
	(0.010)	(0.012)	(0.008)	(0.010)	(0.009)	(0.011)
Flash Flood × High Slack	-0.285	-0.445*				
(=1 if Past Qtr. Unemp. $>$ 90th perc.)	(0.176)	(0.244)				
Non-Flash Flood × High Slack	-0.095*	-0.112**				
(=1 if Past Qtr. Unemp. $>$ 90th perc.)	(0.051)	(0.054)				
Flash Flood × High Slack			-0.738***	-1.07***		
(=1 if *all* Past 4 Qtrs. Unemp. $>$ 90th perc.)			(0.244)	(0.343)		
Non-Flash Flood × High Slack			-0.189***	-0.199***		
(=1 if *all* Past 4 Qtrs. Unemp. $>$ 90th perc.)			(0.067)	(0.072)		
Flash Flood × High Slack					-0.848***	-1.02**
(=1 if *all* Past 8 Qtrs. Unemp. $>$ 90th perc.)					(0.325)	(0.451)
Non-Flash Flood × High Slack					-0.285***	-0.307***
(=1 if *all* Past 8 Qtrs. Unemp. $>$ 90th perc.)					(0.071)	(0.087)
$FlashFlood_{t-1}$	✓	✓				
$Non-FlashFlood_{t-1}$	✓	✓				
Cumulative Flash Floods- Past 4 Qtrs.			✓	✓		
Cumulative Non-Flash Floods- Past 4 Qtrs.			✓	✓		
Cumulative Flash Floods- Past 8 Qtrs.					✓	✓
Cumulative Non-Flash Floods- Past 8 Qtrs.					✓	✓
Net Effect- Flash Flood in	-0.396**	-0.571**				
High Slack Counties	(0.177)	(0.244)				
Net Effect- Flash Flood in			-0.844***	-1.188***		
High Slack Counties			(0.244)	(0.343)		
Net Effect- Flash Flood in					-0.967***	-1.161***
High Slack Counties					(0.321)	(0.448)
Net Effect- Non-Flash Flood in	-0.095**	-0.111**				
High Slack Counties	(0.048)	(0.051)				
Net Effect- Non-Flash Flood in			-0.192***	-0.204***		
High Slack Counties			(0.064)	(0.069)		
Net Effect- Non-Flash Flood in					-0.281***	-0.302***
High Slack Counties					(0.069)	(0.084)
Observations	342,678	342,678	333,405	333,405	321,013	321,013
St×Qtr×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: In columns (1) and (2) the High Slack indicator variable takes value 1 if in the past quarter the unemployment rate in the county was above the 90th percentile of the national distribution over the same quarter-year, and 0 otherwise. Alternatively for columns (3) and (4), counties are classified as high-slack if unemployment exceeded 90th percentile in each of the past four quarters, and finally in columns (5) and (6) counties are classified as high-slack if unemployment exceeded 90th percentile in each of the past eight quarters. Controls include average precipitation and average temperature. Additionally, we control for past quarter flash floods and non-flash floods in columns (1) and (2), cumulative flash flood and non-flash flood exposures in past-4 quarters in columns (3) and (4) and cumulative flash flood and non-flash flood exposures in past-8 quarters in columns (5) and (6). All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A5: Coastal vs. Inland: Differences in Share of Quarters with High-Slack Status

Variable	Mean (Inland)	Mean (Coastal)	Diff	p-value	Sig.
Past qtr. unemployment above 90th percentile	0.10	0.11	0.01	0.372	
Past 4 qtrs. unemployment above 90th percentile	0.05	0.04	-0.01	0.125	
Past 8 qtrs. unemployment above 90th percentile	0.04	0.02	-0.01	0.040	**

Notes: Entries are means of county-level indicators. Each county value is the fraction of quarters in the panel for which the high-slack indicator equals 1. p-values are from two-sample t-tests comparing coastal to inland counties. Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table A6: Heterogeneous Effect by High-Poverty and Low-Median Income

Dependent Variables:	log(Emp.)	log(Tot.Wages)	log(Emp.)	log(Tot.Wages)
Model:	(1)	(2)	(3)	(4)
Variables				
Flash Flood	-0.131***	-0.121**	-0.119**	-0.106*
	(0.049)	(0.059)	(0.049)	(0.059)
Non-Flash Flood	0.003	0.003	0.003	0.006
	(0.008)	(0.010)	(0.008)	(0.010)
Flash Flood × High Poverty	-0.065	-0.438		
	(0.192)	(0.300)		
Non-Flash Flood × High Poverty	-0.080**	-0.080		
	(0.038)	(0.055)		
Flash Flood × Low Median Income			-0.221	-0.656**
			(0.207)	(0.316)
Non-Flash Flood × Low Median Income			-0.104**	-0.135**
			(0.042)	(0.053)
Cumulative Flash Flood- Past 8 Qtrs.	✓	✓	✓	✓
Cumulative Non-Flash Flood- Past 8 Qtrs.	✓	✓	✓	✓
Net Effect - Flash Flood in	-0.196	-0.559*		
High Poverty Counties	(0.185)	(0.291)		
Net Effect - Non-Flash Flood in	-0.077**	-0.078		
High Poverty Counties	(0.036)	(0.053)		
Net Effect - Flash Flood in			-0.340*	-0.761**
Low Median-Income Counties			(0.200)	(0.306)
Net Effect - Non-Flash Flood in			-0.101**	-0.129***
Low Median-Income Counties			(0.040)	(0.050)
Observations	320,941	320,941	320,941	320,941
St×Qtr×Year FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes

Notes: High Poverty is a time-varying classification (indicator variable=1) based on whether a county's poverty rate (including all ages) in the *previous year* is above the 90th percentile of the national county-level poverty distribution in that year, the indicator variable takes value 0 otherwise. Similarly, Low Median Income is a time-varying classification (indicator variable=1) based on whether a county's median household income in the *previous year* is below the 10th percentile of the national county-level median income distribution in that year, the indicator variable takes value 0 otherwise. Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. Additionally, we control for cumulative flash flood and non-flash flood exposure in the past-8 quarters. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A7: Effect of Flash Flood and Non-Flash Flood Days on Total Private Employment and Wages

Dep. Vars.:	log(Emp.)				log(Tot.Wages)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Flash Flood	-0.126**	-0.105***	-0.069***	-0.077***	-0.153**	-0.111***	-0.099***	-0.102***
	(0.053)	(0.033)	(0.020)	(0.018)	(0.065)	(0.034)	(0.023)	(0.022)
Flood	-0.008	-0.009**	-0.005*	-0.004	-0.008	-0.008	-0.002	-0.002
	(0.011)	(0.004)	(0.003)	(0.002)	(0.014)	(0.005)	(0.004)	(0.004)
Fixed-effects								
Observations	345,797	345,797	345,797	345,797	345,797	345,797	345,797	345,797
St×Qtr×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes				Yes			
County×Year		Yes	Yes	Yes		Yes	Yes	Yes
Sub-Region(HUC4)×Qtr FE			Yes				Yes	
Basin(HUC6)×Qtr FE				Yes				Yes

Notes: HUC Sub-region and basin boundaries are obtained from the USGS National Hydrography Dataset (NHDPlus) (U.S. Geological Survey, 2020). County centroids, constructed using Census TIGER/Line shapefiles (U.S. Census Bureau, 2020), are used to assign each county to its HUC sub-region or basin. Controls include average precipitation and average temperature. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A8: Controlling for Rainfall Anomalies

Dependent Variables:	log(Emp.) (1)	log(Emp.) (2)	log(Emp.) (3)	log(Emp.) (4)	log(Tot. Wages) (5)	log(Tot. Wages) (6)	log(Tot. Wages) (7)	log(Tot. Wages) (8)
Variables								
Flash Flood	-0.127**	-0.128**	-0.125**	-0.122**	-0.154**	-0.153**	-0.151**	-0.147**
	(0.053)	(0.053)	(0.054)	(0.053)	(0.065)	(0.066)	(0.066)	(0.066)
Non-Flash Flood	-0.008	-0.008	-0.008	-0.008	-0.008	-0.008	-0.008	-0.008
	(0.011)	(0.011)	(0.011)	(0.011)	(0.014)	(0.014)	(0.014)	(0.014)
#Days ≥ 1-inch Rain	✓				✓			
#Days ≥ 50mm Rain		✓				✓		
#Days ≥ 75mm Rain			✓				✓	
#Days ≥ 100mm Rain				✓				✓
Observations	345,797	345,797	345,797	345,797	345,797	345,797	345,797	345,797
St×Qtr×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Other controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A9: Effect of Flash Flood and Non-Flash Flood Days on Total Private Employment and Wages (Commuting-Zone (CZ) level analysis)

Dependent Variables:	log(Emp.) (1)	log(Total Wages) (2)
Flash Flood	-0.142***	-0.225***
	(0.040)	(0.052)
Flood	-0.0006	0.002
	(0.010)	(0.014)
Observations	80,740	80,740
Commuting-Zone (CZ) FE	Yes	Yes
Census Division×Qtr×Year FE	Yes	Yes

Notes: Controls include average precipitation and average temperature. All specifications include commuting-zone and census-division×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (CZ- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A10: Effect of Flash Flood and Non-Flash Flood Days on Total Private Employment and Wages (with Spatially robust Conley Standard Errors)

Dependent Variables:	log(Emp.) 25 km (1)	log(Emp.) 50 km (2)	log(Emp.) 75 km (3)	log(Emp.) 100 km (4)	log(Tot. Wages) 25 km (5)	log(Tot. Wages) 50 km (6)	log(Tot. Wages) 75 km (7)	log(Tot. Wages) 100 km (8)
Flash Flood	-0.126**	-0.126**	-0.126**	-0.126*	-0.153**	-0.153**	-0.153**	-0.153*
	(0.053)	(0.058)	(0.064)	(0.070)	(0.065)	(0.070)	(0.078)	(0.085)
Non-Flash Flood	-0.008	-0.008	-0.008	-0.008	-0.008	-0.008	-0.008	-0.008
	(0.012)	(0.011)	(0.013)	(0.014)	(0.014)	(0.014)	(0.016)	(0.016)
Observations	345,797	345,797	345,797	345,797	345,797	345,797	345,797	345,797
St×Qtr×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

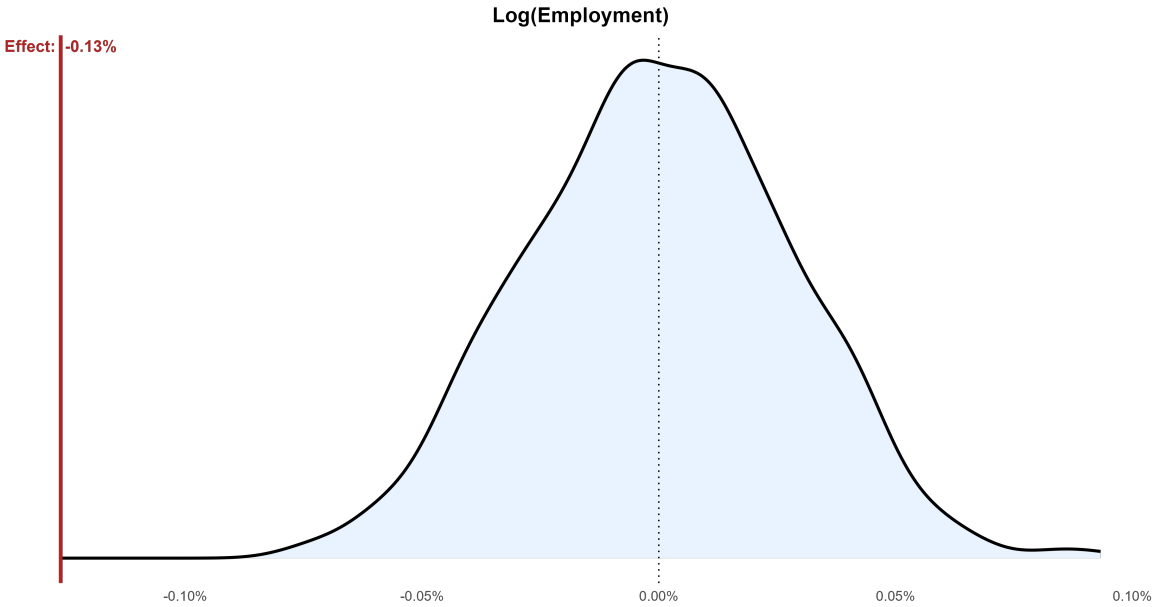
Notes: Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Spatially robust Conley standard errors in parentheses. Columns (1) and (5) use a spatial cutoff of 25 km, (2) and (6) use 50 km, (3) and (7) use 75 km and (4) and (8) use 100 km cutoff. *** p<0.01, ** p<0.05, * p<0.1.

Table A11: Heterogeneous Effect by High Damage Reported Quarters & FEMA Declared Emergency Quarters

Dependent Variables:	log(Emp.) (1)	log(Tot.Wages) (2)	log(Emp.) (3)	log(Tot.Wages) (4)
Variables				
Flash Flood	-0.136**	-0.151**	-0.108**	-0.120*
	(0.055)	(0.068)	(0.053)	(0.066)
Non-Flash Flood	-0.008	-0.008	-0.010	-0.012
	(0.012)	(0.015)	(0.014)	(0.017)
Flash Flood × High Damage Reported	0.079	-0.015		
(=*1 if Reported Damages> 90th perc.*)	(0.068)	(0.086)		
Non-Flash Flood × High Damage Reported	-0.0002	0.0009		
(=*1 if Reported Damages> 90th perc.*)	(0.016)	(0.019)		
Flash Flood × FEMA Declared Emergency			-0.243**	-0.436***
(=*1 if a Qtr. in FEMA Declared Flood Disaster*)			(0.121)	(0.148)
Non-Flash Flood × FEMA Declared Emergency			0.012	0.018
(=*1 if a Qtr. in FEMA Declared Flood Disaster*)			(0.020)	(0.024)
Net Effect - Flash Flood in	-0.057	-0.166*		
High Damage Reported	(0.074)	(0.091)		
Net Effect - Non-Flash Flood in	-0.008	-0.007		
High Damage Reported	(0.016)	(0.019)		
Net Effect- Flash Flood in			-0.351***	-0.556***
FEMA Declared Emergency			(0.128)	(0.154)
Net Effect- Non-Flash Flood in			0.001	0.006
FEMA Declared Emergency			(0.015)	(0.019)
Observations	345,797	345,797	345,797	345,797
St×Qtr×Year FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes

Notes: High-Damage Reported indicator takes value 1 if a county's reported damages in a given year-quarter exceeded 90th percentile of the positive-damages distribution over 1996-2023, and 0 otherwise. FEMA Declared Emergency indicator is set to 1 for county-year-quarters that overlap with a FEMA or Presidentially declared flood disaster period, and 0 otherwise. Controls include average precipitation and average temperature. All specifications include county and state×quarter×year fixed effects. All coefficients and standard errors have been multiplied by 100. Clustered (county- level) standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure A6: Placebo-Analysis- Within County



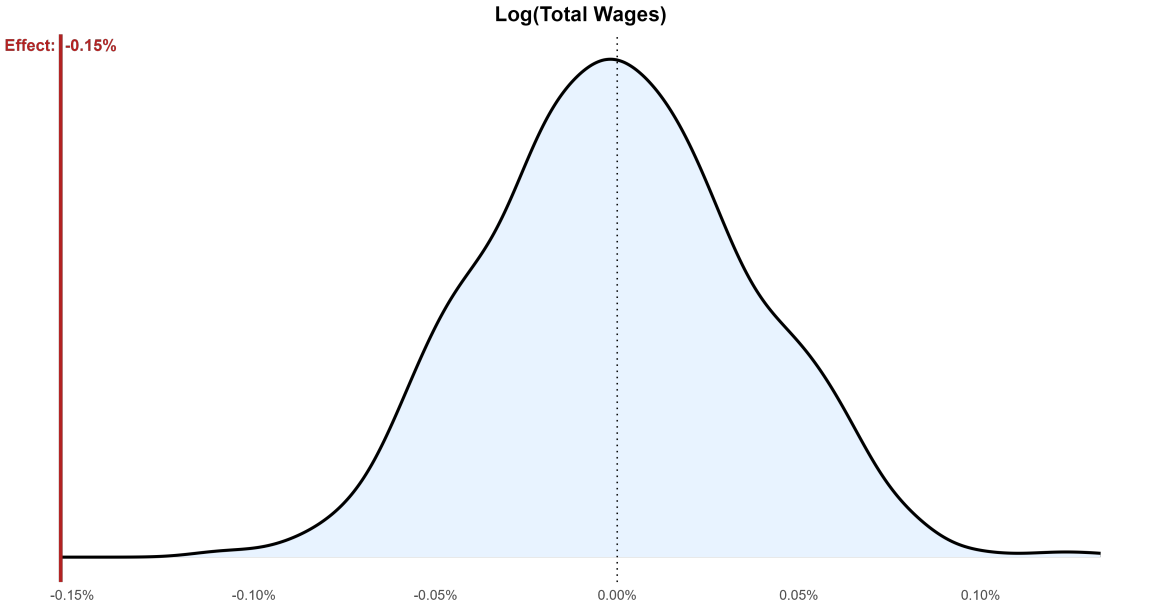


Figure A7: Placebo Analysis- Within State-Quarter-Year

